

Cam contours are usually functions of high degree. When they are wrapped around their base or prime circles, they may have portions which are concave, convex, or flat. Infinitesimally short flats of infinite radius will occur at all inflection points on the cam surface where it changes from concave to convex or vice versa.

The radius of curvature of the finished cam is of concern regardless of the follower type, but the concerns are different for different followers. Figure 8-44 shows an obvious problem with a roller follower whose own (constant) radius of curvature R_f is too large to follow the locally smaller concave (negative) radius $-\rho$ on the cam.

A more subtle problem occurs when the roller follower radius R_f is larger than the smallest positive (convex) local radius $+\rho$ on the cam. This problem is called **undercutting** and is depicted in Figure 8-45. Recall that for a roller follower cam, the cam contour is actually defined as the locus of the center of the roller follower, or the **pitch curve**. The machinist is given these x,y coordinate data (on computer tape or disk) and also told the radius of the follower R_f . The machinist will then cut the cam with a cutter of the same effective radius as the follower, following the pitch curve coordinates with the center of the cutter.

Figure 8-45a shows the situation in which the follower (cutter) radius R_f is at one point exactly equal to the minimum convex radius of curvature of the cam ($+\rho_{min}$). The cutter creates a perfect sharp point, or **cusp**, on the cam surface. This cam will not run very well at speed! Figure 8-45b shows the situation in which the follower (cutter) radius is greater than the minimum convex radius of curvature of the cam. The cutter now undercuts or removes material needed for cam contours in different locations and also creates a sharp point or cusp on the cam surface. This cam no longer has the same displacement function you so carefully designed.

The rule of thumb is to keep the absolute value of the minimum radius of curvature ρ_{min} of the cam pitch curve preferably at least 2 to 3 times as large as the radius of the roller follower R_f .

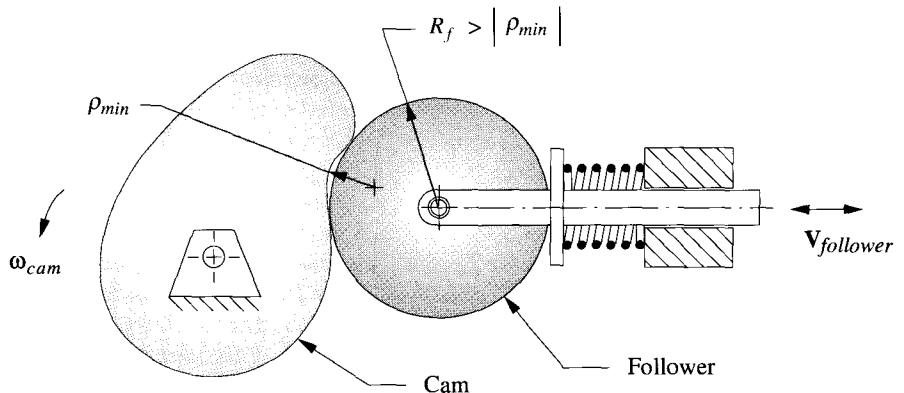


FIGURE 8-44

The result of using a roller follower larger than the one for which the cam was designed

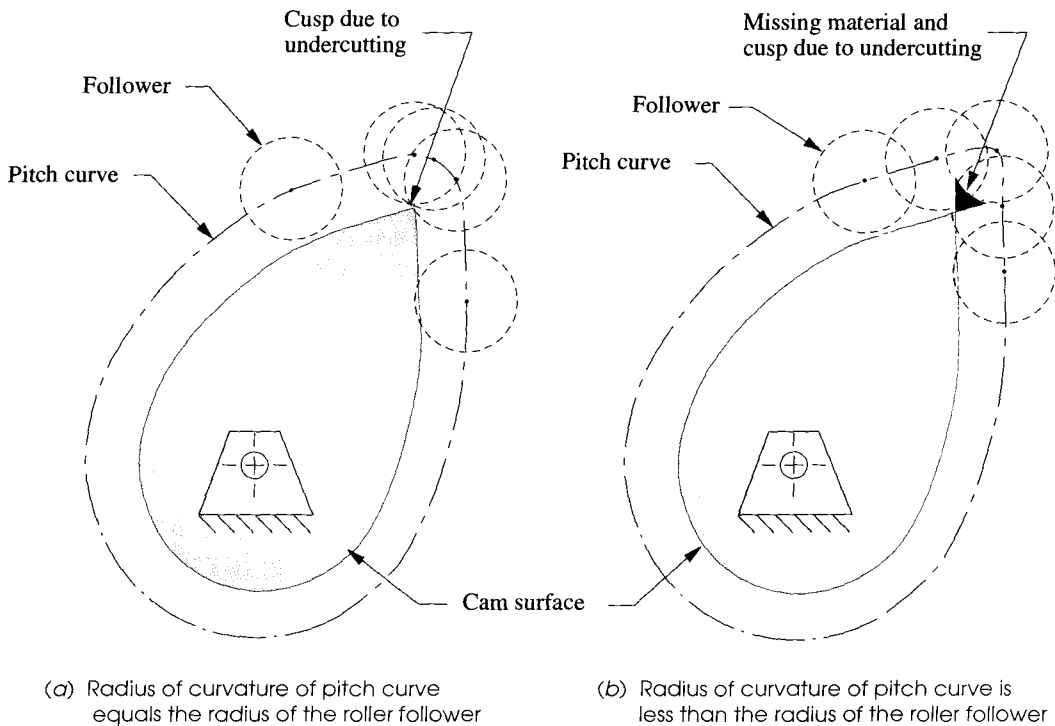


FIGURE 8-45

Small positive radius of curvature can cause undercutting

$$|\rho_{\min}| \gg R_f \quad (8.30)$$

A derivation for radius of curvature can be found in any calculus text. For our case of a roller follower, we can write the equation for the radius of curvature of the pitch curve of the cam as:

$$\rho_{pitch} = \frac{\left[(R_p + s)^2 + v^2 \right]^{3/2}}{(R_p + s)^2 + 2v^2 - a(R_p + s)} \quad (8.31)$$

In this expression, s , v , and a are the displacement, velocity, and acceleration of the cam program as defined in a previous section. Their units are length, length/rad, and length/rad², respectively. R_p is the prime circle radius. **Do not confuse this prime circle radius R_p with the radius of curvature, ρ_{pitch} .** R_p is a **constant value** which you choose as a design parameter and ρ_{pitch} is the constantly changing radius of curvature which results from your design choices.

Also do not confuse R_p , the *prime circle radius* with R_f , the *radius of the roller follower*. See Figure 8-39 (p. 398) for definitions. You can choose the value of R_f to suit the problem, so you might think that it is simple to satisfy equation 8.30 by just selecting

a roller follower with a small value of R_f . Unfortunately it is more complicated than that, as a small roller follower may not be strong enough to withstand the dynamic forces from the cam. The radius of the pin on which the roller follower pivots is substantially smaller than R_f because of the space needed for roller or ball bearings within the follower. Dynamic forces will be addressed in later chapters where we will revisit this problem.

We can solve equation 8.31 for ρ_{pitch} since we know s , v , and a for all values of θ and can choose a trial R_p . If the pressure angle has already been calculated, the R_p found for its acceptable values should be used to calculate ρ_{pitch} as well. If a suitable follower radius cannot be found which satisfies equation 8.30 for the minimum values of ρ_{pitch} calculated from equation 8.31, then further iteration will be needed, possibly including a redefinition of the cam specifications.

Program DYNACAM calculates ρ_{pitch} for all values of θ for a user supplied prime circle radius R_p . Figure 8-46 shows ρ_{pitch} for the four-dwell cam of Figure 8-6 (p. 353). Note that this cam has both positive and negative radii of curvature. The large values of radius of curvature are truncated at arbitrary levels on the plot as they are heading to infinity at the inflection points between convex and concave portions. Note that the radii of curvature go out to positive infinity and return from negative infinity or vice versa at these inflection points (perhaps after a round trip through the universe?).

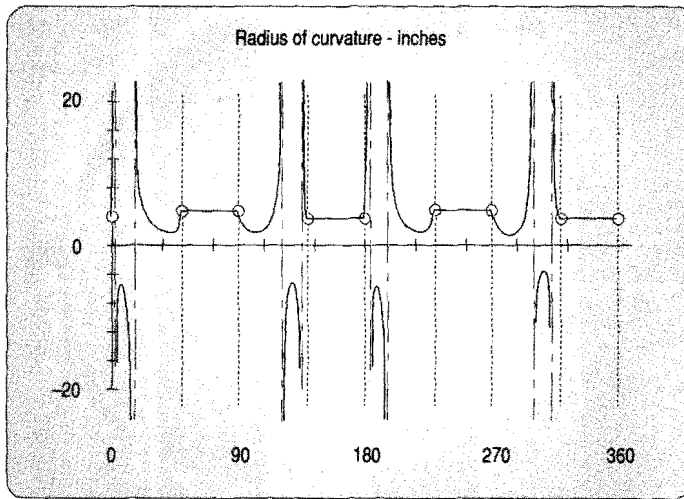
Once an acceptable prime circle radius and roller follower radius are determined based on pressure angle and radius of curvature considerations, the cam can be drawn in finished form and subsequently manufactured. Figure 8-47 shows the profile of the four dwell cam from Figure 8-6. The cam surface contour is swept out by the envelope of follower positions just as the cutter will create the cam in metal. The sidebar shows the parameters for the design, which is an acceptable one. The ρ_{min} is 1.7 times R_f and the pressure angles are less than 30° . The contours on the cam surface appear smooth, with no sharp corners. Figure 8-48 shows the same cam with only one change. The radius of follower R_f has been made the same as the minimum radius of curvature, ρ_{min} . The sharp corners or cusps in several places indicate that undercutting has occurred. This has now become an **unacceptable cam**, *simply because of a roller follower that is too large*.

The coordinates for the cam contour, measured to the locus of the center of the roller follower, or the **pitch curve** as shown in Figure 8-47, are defined by the following expressions, referenced to the center of rotation of the cam. See Figure 8-42 (p. 402) for nomenclature. The subtraction of the cam input angle θ from 2π is necessary because the relative motion of the follower versus the cam is opposite to that of the cam versus the follower. In other words, to define the contour of the centerline of the follower's path around a stationary cam, we must move the follower (and also the cutter to make the cam) in the opposite direction of cam rotation.

$$\begin{aligned}x &= \cos \lambda \sqrt{(d+s)^2 + \varepsilon^2} \\y &= \sin \lambda \sqrt{(d+s)^2 + \varepsilon^2}\end{aligned}\tag{8.32}$$

where :

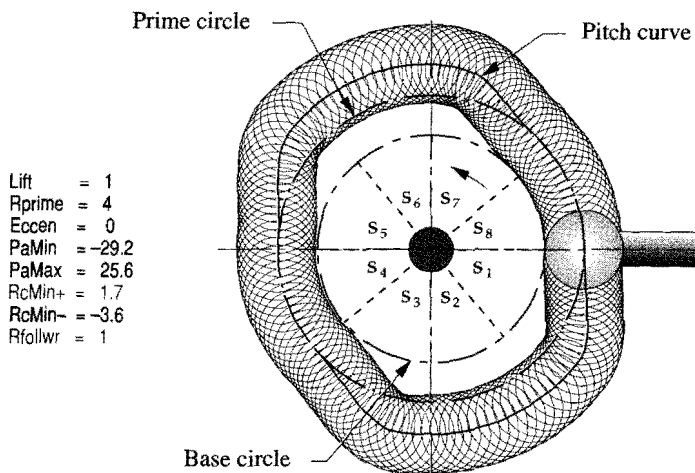
$$\lambda = (2\pi - \theta) - \arctan\left(\frac{\varepsilon}{d+s}\right)$$

**FIGURE 8-46**

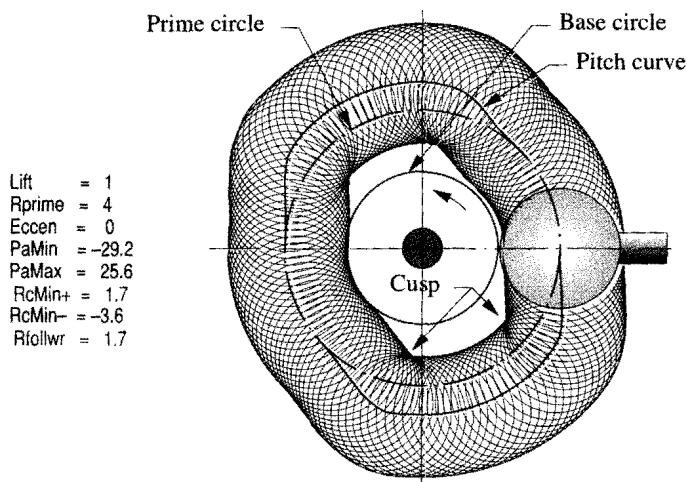
Radius of curvature of a four-dwell cam

Radius of Curvature—Flat-Faced Follower

The situation with a flat-faced follower is different to that of a roller follower. A negative radius of curvature on the cam cannot be accommodated with a flat-faced follower. The flat follower obviously cannot follow a concave cam. Undercutting will occur when the radius of curvature becomes negative if a cam with that condition is made.

**FIGURE 8-47**

Radial plate cam profile is generated by the locus of the roller follower (or cutter)

**FIGURE 8-48**

Cusps formed by undercutting due to radius of follower $R_f \geq$ cam radius of curvature ρ

Figure 8-49 shows a cam and flat-faced follower in an arbitrary position. The origin of the global XY coordinate system is placed at the cam's center of rotation, and the X axis is defined parallel to the common tangent, which is the surface of the flat follower. The vector $\mathbf{r}$ is attached to the cam, rotates with it, and serves as the reference line to which the cam angle θ is measured from the X axis. The point of contact A is defined by the position vector $\mathbf{R}_A$. The instantaneous center of curvature is at C and the radius of curvature is ρ . R_b is the radius of the base circle and s is the displacement of the follower for angle θ . The eccentricity is ϵ .

We can define the location of contact point A from two vector loops (in complex notation).

$$\mathbf{R}_A = x + j(R_b + s)$$

and

$$\mathbf{R}_A = ce^{j(\theta+\alpha)} + j\rho$$

so :

$$ce^{j(\theta+\alpha)} + j\rho = x + j(R_b + s) \quad (8.33a)$$

Substitute the Euler equivalent (equation 4.4a, p. 155) in equation 8.33a and separate the real and imaginary parts.

real:

$$c \cos(\theta + \alpha) = x \quad (8.33b)$$

imaginary:

$$c \sin(\theta + \alpha) + \rho = R_b + s \quad (8.33c)$$

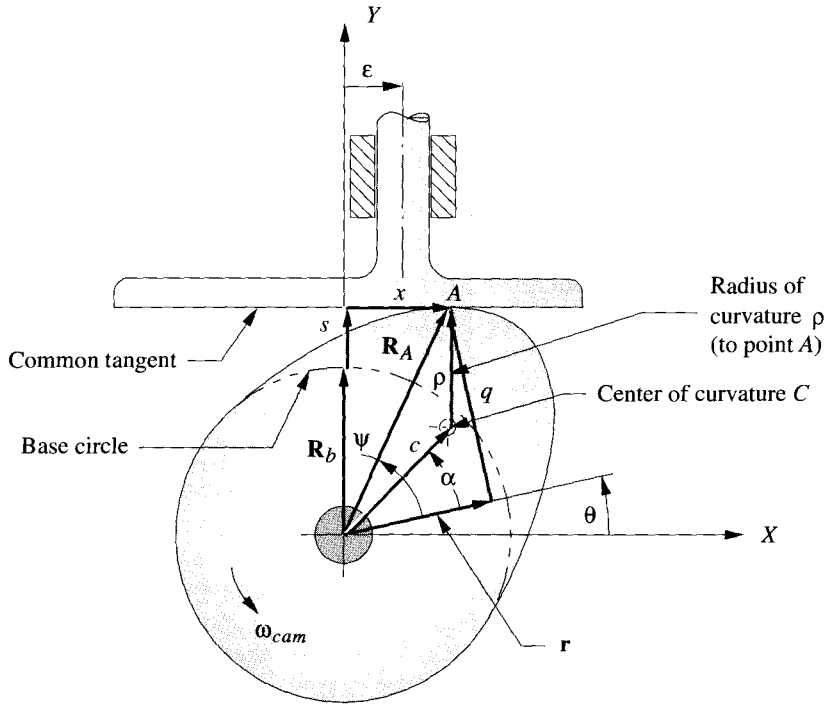


FIGURE 8-49

Geometry for derivation of radius of curvature and cam contour with flat-faced follower

The center of curvature C is **stationary** on the cam meaning that the magnitudes of c and ρ , and angle α do not change for small changes in cam angle θ . (These values are not constant but are at stationary values. Their first derivatives with respect to θ are zero, but their higher derivatives are not zero.)

Differentiating equation 8.33a with respect to θ then gives:

$$jce^{j(\theta+\alpha)} = \frac{dx}{d\theta} + j \frac{ds}{d\theta} \quad (8.34)$$

Substitute the Euler equivalent (equation 4.4a, p. 155) in equation 8.34 and separate the real and imaginary parts.

real:

$$-c \sin(\theta + \alpha) = \frac{dx}{d\theta} \quad (8.35)$$

imaginary:

$$c \cos(\theta + \alpha) = \frac{ds}{d\theta} = v \quad (8.36)$$

Inspection of equations 8.33b and 8.36 shows that:

$$x = v \quad (8.37)$$

This is an interesting relationship that says the x position of the contact point between cam and follower is numerically equal to the velocity of the follower in length/rad. This means that the v diagram gives a direct measure of the necessary minimum face width of the flat follower.

$$\text{facewidth} > v_{\max} - v_{\min} \quad (8.38)$$

If the velocity function is asymmetric, then a minimum-width follower will have to be asymmetric also, in order not to fall off the cam.

Differentiating equation 8.37 with respect to θ gives:

$$\frac{dx}{d\theta} = \frac{dv}{d\theta} = a \quad (8.39)$$

Equations 8.33c and 8.35 can be solved simultaneously and equation 8.39 substituted in the result to yield:

$$\rho = R_b + s + a \quad (8.40)$$

BASE CIRCLE Note that equation 8.40 defines the radius of curvature in terms of the base circle radius and the displacement and acceleration functions from the s v a j diagrams only. Because ρ cannot be allowed to become negative with a flat-faced follower, we can formulate a relationship from this equation which will predict the minimum base circle radius R_b needed to avoid undercutting. The only factor on the right side of equation 8.40 which can be negative is the acceleration, a . We have defined s to be always positive, as is R_b . Therefore, the worst case for undercutting will occur when a is at its **largest negative value**, $a_{\min}$, whose value we know from the a diagram. The minimum base circle radius can then be defined as:

$$R_{b_{\min}} > \rho_{\min} - s_{@a_{\min}} - a_{\min} \quad (8.41)$$

Note that the value of s in this equation is taken at the cam angle θ corresponding to that of $a_{\min}$. Because the value of $a_{\min}$ is negative and it is also negated in equation 8.41, it dominates the expression. To use this relationship, we must choose some minimum radius of curvature $\rho_{\min}$ for the cam surface as a design parameter. Since the hertzian contact stresses at the contact point are a function of local radius of curvature, that criterion can be used to select $\rho_{\min}$. That topic is beyond the scope of this text and will not be further explored here. See reference 1 for further information on contact stresses.

CAM CONTOUR For a flat-faced follower cam, the coordinates of the physical cam surface must be provided to the machinist as there is no pitch curve to work to. Figure 8-49 (p. 409) shows two orthogonal vectors, $\mathbf{r}$ and $\mathbf{q}$, which define the cartesian coordinates of contact point A between cam and follower with respect to a rotating axis coordinate system embedded in the cam. Vector $\mathbf{r}$ is the rotating “ x ” axis of this embedded coordinate system. Angle ψ defines the position of vector $\mathbf{R}_A$ in this system. Two vector loop equations can be written and equated to define the coordinates of all points on the cam surface as a function of cam angle θ .

$$\mathbf{R}_A = x + j(R_b + s)$$

and

$$\mathbf{R}_A = re^{j\theta} + qe^{j\left(\theta + \frac{\pi}{2}\right)}$$

so:

$$re^{j\theta} + qe^{j\left(\theta + \frac{\pi}{2}\right)} = x + j(R_b + s) \quad (8.42)$$

Divide both sides by $e^{j\theta}$:

$$r + jq = xe^{-j\theta} + j(R_b + s)e^{-j\theta} \quad (8.43)$$

Separate into real and imaginary components and substitute v for x from equation 8.37:

real (x component):

$$r = (R_b + s)\sin\theta + v\cos\theta \quad (8.44a)$$

imaginary (y component):

$$q = (R_b + s)\cos\theta - v\sin\theta \quad (8.44b)$$

Equations 8.44 can be used to machine the cam for a flat-faced follower. These x, y components are in the rotating coordinate system that is embedded in the cam.

Note that none of the equations developed above for this case involve the **eccentricity**, ϵ . It is only a factor in cam size when a roller follower is used. It does not affect the geometry of a flat follower cam.

Figure 8-50 shows the result of trying to use a flat-faced follower on a cam with negative radius of curvature. If the follower could contact the cam at all the points necessary to control the follower to the function in the s diagram, the cam surface would be as developed by the envelope of straight lines. However, these loci of the follower face are cutting into cam contours which are needed for other cam angles. The line running through the forest of follower loci is the theoretical cam contour needed for this design. The undercutting can be clearly seen as the crescent-shaped missing pieces at four places between the cam contour and the follower loci.

SUMMARY The task of sizing a cam is an excellent example of the need for and value of iteration in design. Rapid recalculation of the relevant equations with a tool such as program DYNACAM makes it possible to quickly and painlessly arrive at an acceptable solution while balancing the often conflicting requirements of pressure angle and radius of curvature constraints. In any cam, either the pressure angle or radius of curvature considerations will dictate the minimum size of the cam. Both factors must be checked. The choice of follower type, either roller or flat-faced, makes a big difference in the cam geometry. Cam programs which generate negative radii of curvature are unsuited to the flat-faced type of follower unless very large base circles are used to force p to be positive everywhere.

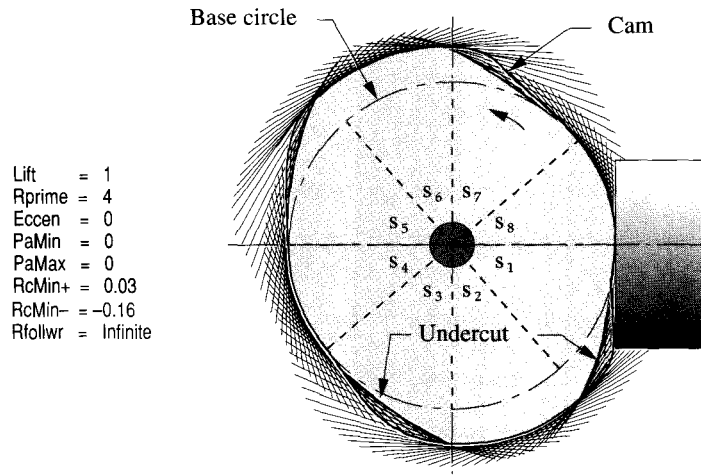


FIGURE 8-50

Undercutting due to negative radius of curvature used with flat-faced follower

8.8 CAM MANUFACTURING CONSIDERATIONS

The preceding sections illustrate that there are a number of factors to consider when designing a cam. A great deal of care in design is necessary to obtain a good compromise of all factors, some of which conflict. Once the cam design is complete a whole new set of considerations must be dealt with that involve manufacturing the cam. After all, if your design cannot be successfully machined in metal in a way that truly represents the theoretical functions chosen, their benefits will not be realized. Unlike linkages, which are very easy to make, cams are a challenge to manufacture properly.

Cams are usually made from strong, hard materials such as **medium to high carbon steels** (case- or through-hardened) or **cast ductile iron or grey cast iron** (case-hardened). Cams for low loads and speeds or marine applications are sometimes made of brass or bronze. Even plastic cams are used in such applications as washing machine timers where the cam is merely tripping a switch at the right time. We will concentrate on the higher load-speed situations here for which steel or cast/ductile iron are the only practical choices. These materials range from fairly difficult to very difficult to machine depending on the alloy. At a minimum, a reasonably accurate milling machine is needed to make a cam. A computer controlled machining center is far preferable and is most often the choice for serious cam production.

Cams are typically milled with rotating cutters that in effect "tear" the metal away leaving a less than perfectly smooth surface at a microscopic level. For a better finish and better geometric accuracy, the cam can be ground after milling away most of the unneeded material. Heat treatment is usually necessary to get sufficient hardness to prevent rapid wear. Steel cams are typically hardened to about Rockwell Rc 50-55. Heat treatment introduces some geometric distortion. The grinding is usually done after heat

treatment to correct the contour as well as to improve the finish.* The grinding step nearly doubles the cost of an already expensive part, so it is often skipped in order to save money. A hardened but unground cam will have some heat distortion error despite accurate milling before hardening. There are several methods of cam manufacture in common use as shown in Table 8-3.

Geometric Generation

Geometric generation refers to the continuous "sweeping out" of a surface as in turning a cylinder on a lathe. This is perhaps the ideal way to make a cam because it creates a truly continuous surface with an accuracy limited only by the quality of the machine and tools used. Unfortunately there are very few types of cams that can be made by this method. The most obvious one is the eccentric cam (Figure 8-10, p. 359) which can be turned and ground on a lathe. A cycloid can also be geometrically generated. Very few other curves can. The presence of dwells makes it extremely difficult to apply this method. Thus, it is seldom used for cams. However, when it can be, as in the case of the eccentric cam of Figure 8-10, the resulting acceleration, though not perfect, is very close to the theoretical cosine wave as seen in Figure 8-11 (p. 359). This eccentric cam was made by turning and grinding on a high-quality lathe. This is the best that can be obtained in cam manufacture. Note that the displacement function is virtually perfect. The errors are only visible in the more sensitive acceleration function measurement.

Manual or NC Machining to Cam Coordinates (Plunge-Cutting)

Computer-aided manufacturing (CAM) has become the virtual standard for high accuracy machining in the United States. Numerical control (NC) machinery comes in many types. Lathes, milling machines, grinders, etc., are all available with on-board computers which control either the position of the workpiece, the tool, or both. The simplest type of NC machine moves the tool (or workpiece) to a specified x,y location and then drives the tool (say a drill) down through the workpiece to make a hole. This process is repeated as much as necessary to create the part. This simple process is referred to as NC to distinguish it from continuous numerical control (CNC).

This NC process is sometimes used for cam manufacture, and even for master cams as described below. It is, in fact, merely a computerized version of the old manual method of cam milling, which is often called **plunge-cutting** to refer to plunging the spinning milling cutter down through the workpiece. This is not the best way to machine a cam because it leaves "scallops" on the surface as shown in Figure 8-51, due to the fact

* Some automotive camshafts are ground soft, then hardened and polished to remove the carburization film from the hardening process. The reason for this is to avoid the possibility of "burning" the surface during grinding which would locally anneal the material and lead to premature surface fatigue failure.

TABLE 8-3 Common Cam Manufacturing Methods

1	Geometric Generation
2	Manual or N umerical C ontrol (NC) machining to cam coordinates (plunge-cutting)
3	C ontinuous N umerical C ontrol (CNC) with L inear I nterpolation (LI)
4	C ontinuous N umerical C ontrol (CNC) with C ircular I nterpolation (CI)
5	A nalog D uplication (AD) of a hand-dressed master cam

that the machinist can only *plunge* at a discrete number of positions around the cam. In effect, the displacement function which we developed has to be “discretized” or sampled at some finite number of places around the cam. Practicality limits this digitizing process to increments of about $1/2$ to 1 degree. With an NC process the increment might be reduced to $1/4$ or $1/10$ degree. At some point “diminishing returns” will set in as the machine’s ability to resolve positions spaced too closely will limit the accuracy. Standard milling machines can be expected to give accuracies in the 0.001 inch tolerance range. Tooling-quality machining centers, jig borers, and grinders can be as much as two to 20 times that accurate [0.0005 -in down to 0.000050 -in (50 millionths) tolerance].

The scallops that are left on the cam after plunge-cutting have to be removed by hand-dressing with files and grindstones. This obviously introduces more error. Even if the bottoms of the scallops at the sample increments were exactly correct, all points between are subject to the vagaries of hand work. The chance of exactly achieving the designed $s v a j$ functions, especially the higher derivatives, with this manufacturing method is slight.

Continuous Numerical Control with Linear Interpolation

In a CNC machine, the tool is in constant contact with the workpiece, always cutting, while the computer controls the movement of the workpiece from position to position as stored in its memory. This is a **continuous cutting process** as opposed to the discrete one of NC. However, the cam displacement function must still be discretized or sampled at some angular increment. The common increments are $1/4$, $1/2$, and 1 degree. Since the machine only has information about the x, y locations of these 360, 720, or 1440 points around the cam, it has to figure out how to get from one point to another while cutting. The most common method used to “fill in” the missing data is **linear interpolation (LI)**. The machine’s computer calculates the straight line between each pair of data points and then drives the cutter (or workpiece) so as to stay as close to that line as it can. If it could do this perfectly (which it can’t), we would get a piecewise continuous first-order approximation to the cam contour. This would introduce slope discontinuities which will create theoretically infinite pulses of acceleration. We would be back to

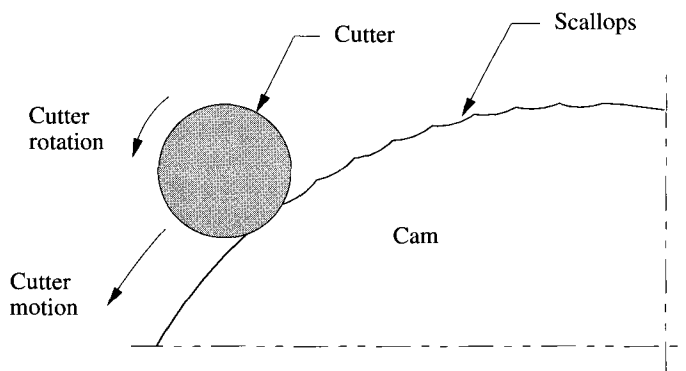


FIGURE 8-51

Plunge-cutting a cam leaves scallops on its surface

our “bad cam by naïve designer” of Example 8-1 having infinite acceleration regardless of the actual function selected.

An improvement can be made in this linear interpolation scheme by fitting a cubic spline curve to the cam coordinate data and then resampling this spline approximation at finer spacing down to the machine’s angular resolution. This denser data set is then used to drive the cutter which still must traverse approximate straight-line paths between the close spaced data points. The curve fitting and resampling is typically done at the manufacturing stage.

Fortunately, the dynamics of the cutting process which are a function of speeds, feeds, tool sharpness, tool chatter, deflection of the spindle, etc., all conspire to prevent the formation of a series of distinct “flats” which would give the derivatives shown in Figure 8-8. Rather, the kind of acceleration curve that actually results from a cam which was milled (but not ground) on a very high-quality CNC machining center, using 1 degree linear interpolation, is as shown in Figure 8-52. The program is a simple harmonic eccentric with no dwells. The dynamic curves were measured with instrumentation on the roller follower while the cam was running at 600 rpm on a custom-designed cam dynamic test fixture (CDTF).^[8] The actual displacement is quite true to the theoretical, but the acceleration has a significant amount of vibratory noise present which distorts the function from its theoretical cosine waveshape. The acceleration is shown in g’s. Compare the peak-to-peak values in Figure 8-52 (8 g) with the same cam design made with geometric generation in Figure 8-11 (p. 359), (5 g). The error is of the order of 3 g’s on a base of 5 g’s. These errors in the acceleration are due to a combination of manufacturing factors as described above. It was found that the use of 1/4, 1/2 or 1 degree digitization increments on this size cam* made no statistically significant difference in the fidelity of the actual acceleration function to its theoretical waveform.^[8]

The physical fidelity of the cam surface of the same 1 degree linear interpolated eccentric cam to a geometrically generated (turned and ground) reference cam can be seen in Figure 8-53 which is an enlarged section of a portion of the cams as measured to

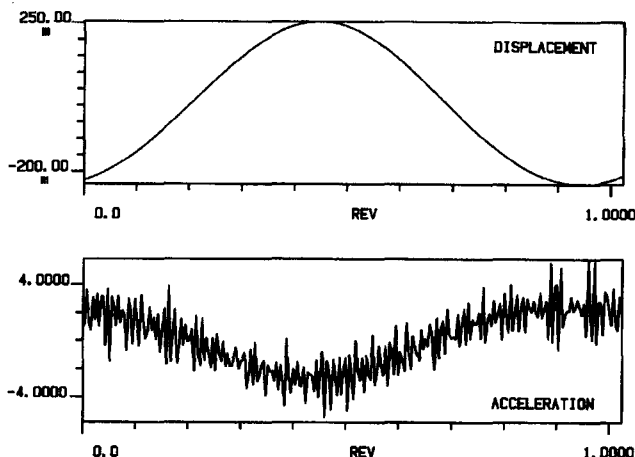


FIGURE 8-52

Displacement and acceleration of eccentric cam made with 1° linear interpolation CNC

* These cams were about 8 in (200 mm) in diameter. If the cam diameter is larger, then smaller angular increments of digitization will be needed as the distance along the pitch curve between data points for any angular increment increases linearly with prime circle diameter.

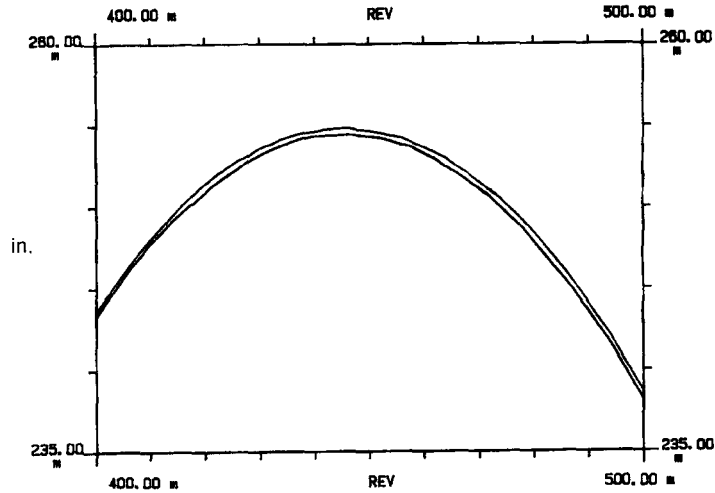


FIGURE 8-53

Contours of 2 cams: one turned and ground; one milled with 1° linear interpolation CNC

0.0005-in accuracy. The axes are calibrated in inches. These figures show that linear interpolated CNC is a reasonably accurate method of cam manufacture.

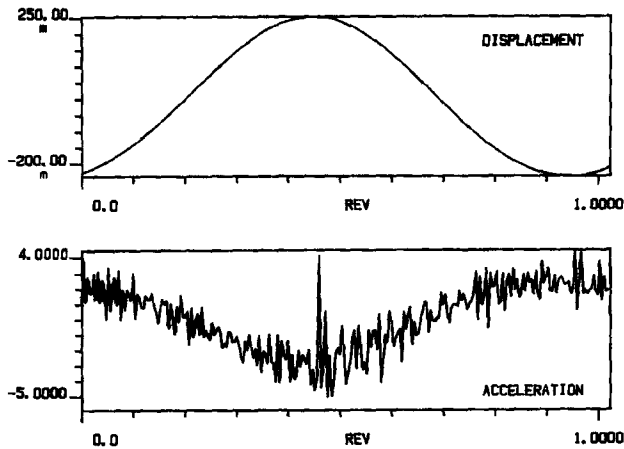
Continuous Numerical Control with Circular Interpolation

This process is similar to CNC with linear interpolation except that a **circular interpolation** (CI) algorithm is used between data points. This potentially allows a sparser (thus smaller) database in the machine which can be an advantage. The computer tries to fit a circle arc to as many adjacent data points as possible without exceeding a user-selected error band around the actual displacement function. This reduces the (variable) number of fitted data points for anyone arc segment to three, a radius and its two center coordinates. Most CNC machines have a built-in algorithm to generate (cut) circle arcs quickly and efficiently, as this is a common requirement in normal machining.

Figure 8-54 shows the same cam design as in Figures 8-11 and 8-52, made on the same machining center from the same bar of steel, but with circular interpolation (CI). The error in acceleration is more than the turned-ground (TG) cam but less than the linear interpolated (LI) one. Figure 8-55 shows the cam contour of the CI cam compared to the TG cam. Based on dynamic performance the circular interpolated cam has lower acceleration error than the linear interpolated cam and the difference is statistically significant. The "blip" in the middle of the period is due to the slight ridge formed at the point where the cutter starts and stops its continuous sweep around the cam contour.

Analog Duplication

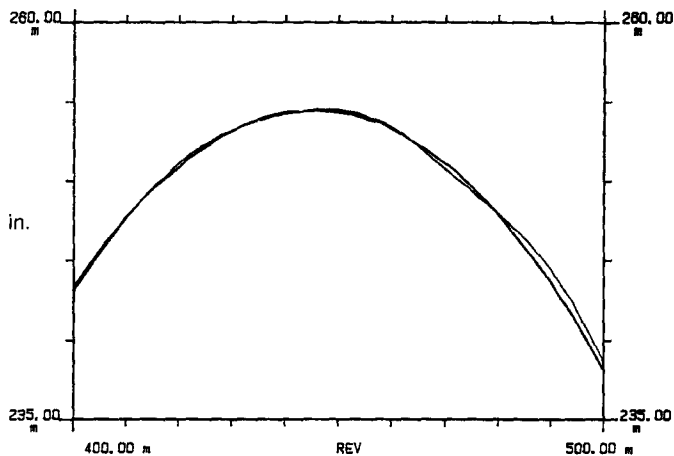
The last method listed, analog duplication, involves the creation of a master cam, sometimes at larger than full scale, which is subsequently used in a cam duplicating machine to turn out large quantities of the finished cams. Some automotive camshafts are still

**FIGURE 8-54**

Displacement and acceleration of eccentric cam made with circular interpolation CNC

made by this method though CNC methods are replacing the automotive cam duplicating machines. Analog duplication can be the most economical method where production quantities are high.

A cam duplicating machine has two spindles. The master cam is mounted on one, and the workpiece is placed on the other. A dummy cutter mounted on one crank of a pantograph linkage is used as a follower against the master. The actual cutter is mounted on the other crank of the pantograph linkage. A multiplication ratio can be introduced

**FIGURE 8-55**

Contours of 2 cams: one turned and ground; one milled with circular interpolation CNC

between the dummy follower and the cutter to size the finished cam versus the master. This allows master cams of larger size to be used which increases the accuracy. As the master and slave slowly turn together, synchronous and in phase, the dummy cutter follows the master's contour and the workpiece is cut to match. This process can be done with either milling (cutting) or grinding of the cam surface. Typically the cam is rough cut first, and then heat treated and reground to finished size. Some cams are left as-milled with no post heat-treat grinding.

This analog duplication method can obviously create a cam that is only as good as the master cam at best. Some errors will be introduced in the duplicating process due to deflections of the tool or machine parts, but the quality of the master cam ultimately limits the quality of the finished cams. The master cam is typically made by one of the other methods listed in Table 8-3 (p. 413), each of which has its limitations. The master cam may require some hand-dressing with files or hand grindstones to smooth its surface. A plunge-cut cam requires a lot of hand-dressing, the CNC cams less so. If hand-dressing is done, it will result in a very smooth surface but the chances that the resulting contour is an accurate representation of the designed $s v a j$ functions, especially the higher derivatives, is slim. Thus the finished cam may not be an accurate representation of the design.

Figure 8-56 shows the same cam design as in Figures 8-52 (p. 415) and 8-54 (p. 417), made on the same machining center from the same bar of steel, but analog duplicated from a hand-dressed, plunge-cut master. This represents the worst case in terms of manufacturing error. The error in acceleration is more than any of the other cams. Figure 8-57 shows the cam contour of the analog milled cam compared to the reference turned-ground cam. It is much less accurate than either of the CNC versions. Based on dynamic performance, the analog milled cam from a hand-dressed, plunge-cut master has a higher acceleration error than any other cam tested and the difference is statistically significant. If the master cam were made by a more accurate method, the accuracy of the production cam could be better but would still be potentially inferior to one made with direct CNC.

Actual Cam Performance Compared to Theoretical Performance

The relative peak accelerations of several common double-dwell cam functions were discussed in Section 8.3 (p. 353) and summarized in Table 8-2 (p. 370). That discussion also emphasized the importance of a smooth jerk function for minimizing vibrations. The theoretical differences between peak accelerations of different cam functions will be altered by the presence of vibratory noise in the actual acceleration waveforms. This noise will be due in part to errors introduced in the manufacturing process, as discussed above, but there will also be inherent differences due to the degree to which the jerk function excites vibrations in the cam-follower train. These vibrations will be heavily influenced by the structural dynamic characteristics of the follower train itself. In general, a very stiff and massive follower train will vibrate less than a light and flexible one, but the presence of frequencies in the cam function that are near the natural frequencies of the follower train will exacerbate the problem.

Figure 8-58 shows the actual acceleration waveforms of four common double-dwell cam programs, modified trapezoid, modified sine, cycloidal, and 4-5-6-7 polynomial, which were ground on the same four-dwell cam. These waveforms were measured with

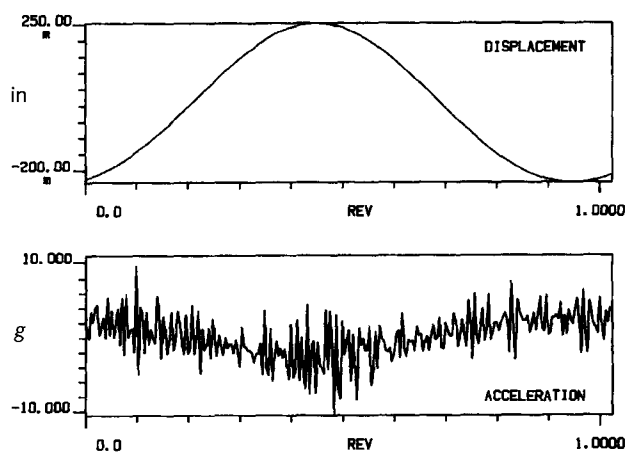


FIGURE 8-56
Displacement and acceleration of eccentric cam made with analog duplication of a hand-dressed master cam

an accelerometer mounted on the stiff follower train of a cam dynamic test fixture (CDTF) specially designed for low vibration and noise. These test cams were made by a specialty cam manufacturer using linear interpolation CNC with a 1/4 degree digitizing increment on very high quality machining and grinding equipment. After hardening, the cam contour was ground to a surface finish with an average roughness (R_a) of 0.125 microns. Comparing these functions on the same cam, made on the same production machinery, and run against the same follower system gives a means to measure their actual differences in relative accelerations.

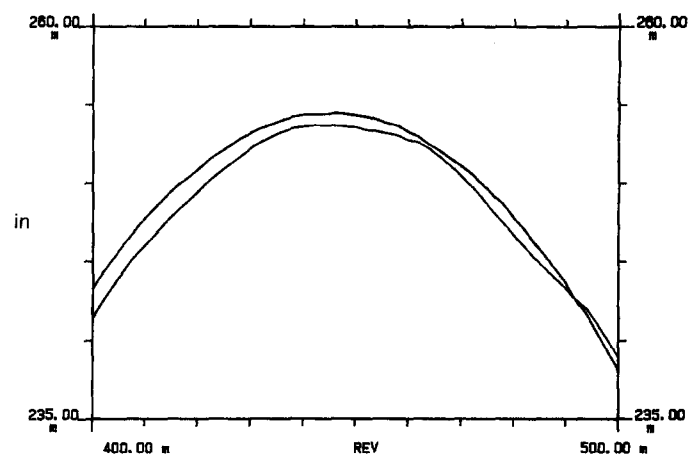


FIGURE 8-57
Contours of 2 cams: one turned and ground; one milled with analog duplication of a hand-dressed master cam

All four acceleration waveforms have some amount of vibratory noise present which is seen as oscillations or ripples on the curves. Compare these curves to their exact theoretical equivalents in Figure 8-18 (p. 370). Table 8-4 compares the expected theoretical peak-to-peak acceleration $A_{pp\text{theoretical}}$ for each cam (column 2) with its actual, measured peak-to-peak acceleration $A_{pp\text{actual}}$ (column 3). The error is expressed in column 4 as an acceleration multiplier factor A_{mf} defined as:

$$A_{mf} = \frac{A_{pp\text{actual}}}{A_{pp\text{theoretical}}} \quad (8.45)$$

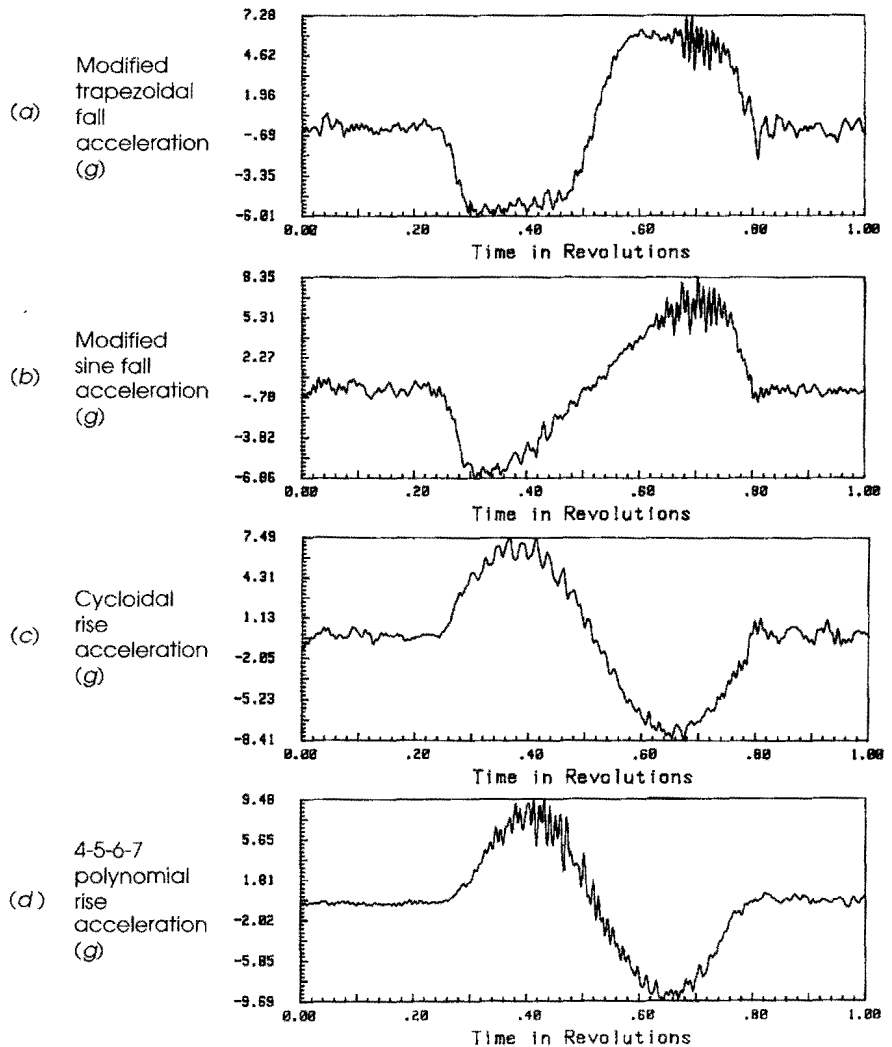


FIGURE 8-58

Actual measured follower acceleration curves of four, double-dwell programs

TABLE 8-4 Actual Peak Acceleration Compared to the Theoretical
for Various Double-Dwell Cam Programs (*g*)

Cam Program	Theor. <i>A_{pp}</i> , <i>g</i>	Actual <i>A_{pp}</i> , <i>g</i>	Ratio <i>A_{mf}</i> (Eq. 8.45)	Theor. * Accel. Factor	Actual Accel. Factor
Mod. trap. accel.	11.78	13.29	1.13	4.888 <i>h/β</i> ²	5.514 <i>h/β</i> ²
Mod. sine. accel.	13.32	15.21	1.14	5.528 <i>h/β</i> ²	6.313 <i>h/β</i> ²
Cycloidal disp.	15.14	15.90	1.05	6.283 <i>h/β</i> ²	6.597 <i>h/β</i> ²
4-5-6-7 poly. disp.	18.14	19.17	1.06	7.526 <i>h/β</i> ²	7.954 <i>h/β</i> ²

* From Table 8-2 (p. 370)

Column five in Table 8-4 shows the maximum acceleration factors for these four functions taken from Table 8-2. The last (sixth) column shows the actual maximum acceleration values based on these test data and is the product of columns four and five.

These values of actual maximum acceleration show smaller differences between the functions than are predicted by their theoretical waveforms. The modified trapezoid, which has the lowest theoretical acceleration, has a 13% noise penalty due to vibration. The modified sine has a 14% noise penalty, while the cycloidal and 4-5-6-7 polynomial functions have only 5 to 6% noise. This is due to the fact that the last two functions have smoother jerk waveforms than the first two. The cycloidal waveform, with its cosine jerk function, is a good choice for high-speed operations. Its acceleration is actually only about 19% greater than the modified trapezoid's and 5% more than the modified sine, rather than the 28% and 14% differentials predicted by the theoretical peak values.

8.9 PRACTICAL DESIGN CONSIDERATIONS

The cam designer is often faced with many confusing decisions, especially at an early stage of the design process. Many early decisions, often made somewhat arbitrarily and without much thought, can have significant and costly consequences later in the design. The following is a discussion of some of the trade-offs involved with such decisions in the hope that it will provide the cam designer with some guidance in making these decisions.

Translating or Oscillating Follower?

There are many cases, especially early in a design, when either translating or rotating motion could be accommodated as output from the cam. Approximate straight-line motion is often adequate and can be obtained from a large-radius rocker follower. The rocker or oscillating follower has one significant advantage over the translating follower when a roller follower is used. A translating follower is free to rotate about its axis of translation and may need to have some antirotation guiding (such as a keyway) provided to prevent misalignment of a roller follower with the cam.

Conversely, the oscillating follower will keep the roller follower aligned in the same plane as the cam with no guiding beyond its own pivot. Also, the pivot friction in an oscillating follower typically has a small moment arm compared to the moment of the force from the cam on the follower arm. But, the friction force on a translating follower has a one-to-one geometric relationship with the cam force. This can have a larger parasitic effect on the system.

On the other hand, translating flat-faced followers are often deliberately arranged with their axis slightly out of the plane of the cam in order to create a rotation about their own axis due to the frictional moment resulting from the offset. The flat follower will then precess around its own axis and distribute the wear over its entire face surface. This is common practice in automotive valve cams that use flat-faced followers or "tappets."

Force or Form-Closed?

A form-closed (track or groove) cam is more expensive to make than a force-closed (open) cam simply because there are two surfaces to machine and grind. Also, heat treating will often distort the track of a form-closed cam, narrowing or widening it such that the roller follower will not fit properly. This virtually requires post heat-treat grinding for track cams in order to resize the slot. An open (force-closed) cam will also distort on heat-treating but may still be usable without grinding.

FOLLOWER JUMP The principal advantage of a form-closed (track) cam is that it does not need a return spring, and thus can be run at higher speeds than a force-closed cam whose spring and follower mass will go into resonance at some speed, causing potentially destructive follower jump. This phenomenon will be investigated in Chapter 15 on cam dynamics. High-speed automobile and motorcycle racing engines often use form-closed (desmodromic) valve cam trains to allow higher engine rpm without incurring valve "float," or follower jump.

CROSSOVER SHOCK Though the lack of a return spring can be an advantage, it comes, as usual, with a trade-off. In a form-closed (track) cam there will be crossover shock each time the acceleration changes sign. Crossover shock describes the impact force that occurs when the follower suddenly jumps from one side of the track to the other as the dynamic force (ma) reverses sign. There is no flexible spring in this system to absorb the force reversal as in the force-closed case. The high impact forces at crossover cause noise, high stresses, and local wear. Also, the roller follower has to reverse direction at each crossover which causes sliding and accelerates follower wear. Studies have shown that roller followers running against a well-lubricated open radial cam have slip rates of less than 1%.^{19]}

Radial or Axial Cam?

This choice is largely dictated by the overall geometry of the machine for which the cam is being designed. If the follower must move parallel to the camshaft axis, then an axial cam is dictated. If there is no such constraint, a radial cam is probably a better choice simply because it is a less complicated, thus cheaper, cam to manufacture.

Roller or Flat-Faced Follower?

The roller follower is a better choice from a cam design standpoint simply because it accepts negative radius of curvature on the cam. This allows more variety in the cam program. Also, for any production quantities, the roller follower has the advantage of being available from several manufacturers in any quantity from one to a million. For low quantities it is not usually economical to design and build your own custom follower. In addition, replacement roller followers can be obtained from suppliers on short notice when repairs are needed. Also, they are not particularly expensive even in small quantities.

Perhaps the largest users of flat-faced followers are automobile engine makers. Their quantities are high enough to allow any custom design they desire. It can be made or purchased economically in large quantity and can be less expensive than a roller follower in that case. Also with engine valve cams, a flat follower can save space over a roller. However, many manufacturers have switched to roller followers in automobile engines to reduce friction and improve fuel mileage. Diesel engines have long used roller followers (tappets) as have racers who "hop-up" engines for high performance.

Cams used in automated production line machinery use stock roller followers almost exclusively. The ability to quickly change a worn follower for a new one taken from the stockroom without losing much production time on the "line" is a strong argument in this environment. Roller followers come in several varieties (see Figure 8-5a, p. 351). They are based on roller or ball bearings. Plain bearing versions are also available for low-noise requirements. The outer surface, which rolls against the cam can be either cylindrical or spherical in shape. The "crown" on the spherical follower is slight, but it guarantees that the follower will ride near the center of a flat cam regardless of the accuracy of alignment of the axes of rotation of cam and follower. If a cylindrical follower is chosen and care is not taken to align the axes of cam and roller follower, the follower will ride on one edge and wear rapidly.

Commercial roller followers are typically made of high carbon alloy steel such as AISI 52100 and hardened to Rockwell Rc 60 - 62. The 52100 alloy is well suited to thin sections that must be heat-treated to a uniform hardness. Because the roller makes many revolutions for each cam rotation, its wear rate may be higher than that of the cam. Chrome plating the follower can markedly improve its life. Chrome is harder than steel at about Rc 70. Steel cams are typically hardened to a range of Rc 50 - 55.

To Dwell or Not to Dwell?

The need for a dwell is usually clear from the problem specifications. If the follower must be held stationary for any time, then a dwell is required. Some cam designers tend to insert dwells in situations where they are not specifically needed for follower stasis, in a mistaken belief that this is preferable to providing a rise-return motion when that is what is really needed. If the designer is attempting to use a double-dwell program in a single-dwell case, then perhaps his or her motivation to "let the vibrations settle out" by providing a "short dwell" at the end of the motion is justified. However, he or she probably should be using another cam program, perhaps a polynomial tailored to the specifications. Taking the acceleration to zero, whether for an instant or for a "short dwell," is generally unnecessary and undesirable. (See Examples 8-5, p. 375, 8-6, p. 377, and 8-8,

p. 383) A dwell should be used only when the follower is required to be stationary for some measurable time. Moreover, if you do not need any dwell at all, consider using a linkage instead. They are a lot easier and cheaper to manufacture.

To Grind or Not to Grind?

Many production machinery cams are used as-milled, and not ground. Automotive valve cams are ground. The reasons are largely due to cost and quantity considerations as well as the high speeds of automotive cams. There is no question that a ground cam is superior to a milled cam. The question in each case is whether the advantage gained is worth the cost. In small quantities, as are typical of production machinery, grinding about doubles the cost of a cam. The advantages in terms of smoothness and quietness of operation, and of wear, are not in the same ratio as the cost difference. A well-machined cam can perform nearly as well as a well-ground cam and better than a poorly ground cam.^[8]

Automotive cams are made in large quantity, run at very high speed, and are expected to last for a very long time with minimal maintenance. This is a very challenging specification. It is a great credit to the engineering of these cams that they very seldom fail in 100,000 miles or more of operation. These cams are made on specialized equipment which keeps the cost of their grinding to a minimum.

To Lubricate or Not to Lubricate?

Cams need lots of lubrication. Automotive cams are literally drowned in a flow of engine oil. Many production machine cams run immersed in an oil bath. These are reasonably happy cams. Others are not so fortunate. Cams which operate in close proximity to the product on an assembly machine in which oil would cause contamination of the product (food products, personal products) often are run dry. Camera mechanisms, which are full of linkages and cams, are often run dry. Lubricant would eventually find its way to the film.

Unless there is some good reason to eschew lubrication, a cam-follower should be provided with a generous supply of clean lubricant, preferably a hypoid-type oil containing additives for boundary lubrication conditions. The geometry of a cam-follower joint (half-joint) is among the worst possible from a lubrication standpoint. Unlike a journal bearing, which tends to trap a film of lubricant within the joint, the half joint is continually trying to squeeze the lubricant out of itself. This can result in a boundary, or mixed boundary/EHD* lubrication state in which some metal-to-metal contact will occur. Lubricant must be continually resupplied to the joint. Another purpose of the liquid lubricant is to remove the heat of friction from the joint. If run dry, significantly higher material temperatures will result, with accelerated wear and possible early failure.

8.10 REFERENCES

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* EHD = ElastoHydroDynamic.

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8.11 PROBLEMS

Programs DYNACAM and MATRIX may be used to solve these problems or to check your solution where appropriate.

- *8-1 Figure P8-1 shows the cam and follower from Problem 6-65. Using graphical methods, find and sketch the equivalent fourbar linkage for this position of the cam and follower.
- 8-2 Figure P8-1 shows the cam and follower from Problem 6-65. Using graphical methods, find the pressure angle at the position shown.
- 8-3 Figure P8-2 shows a cam and follower. Using graphical methods, find and sketch the equivalent fourbar linkage for this position of the cam and follower.

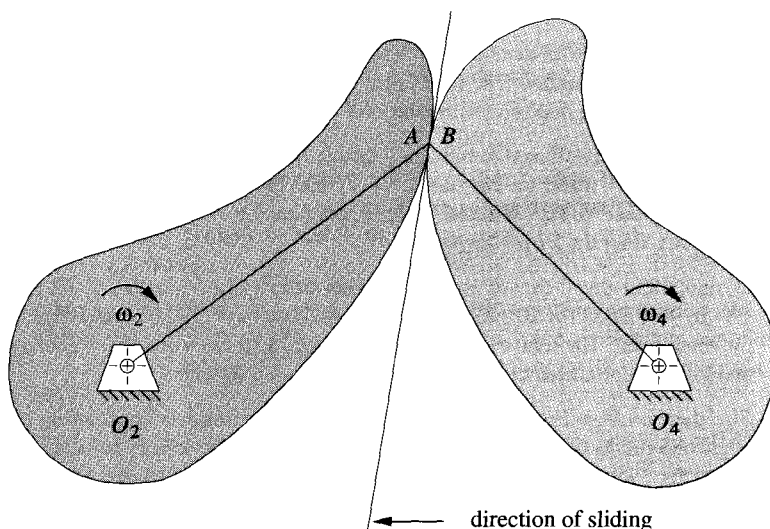
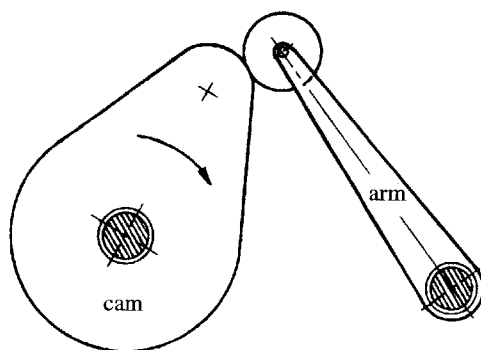


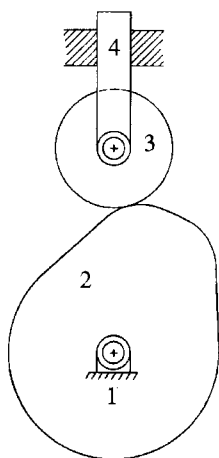
FIGURE P8-1

Problems 8-1 to 8-2. Adapted from P. H. Hill and W. P. Rule, (1960). *Mechanisms: Analysis and Design*.

* Answers in Appendix F.

**FIGURE P8-2**

Problems 8-3 to 8-4

**FIGURE P8-3**

Problems 8-5 to 8-6

- 8-4 Figure P8-2 shows a cam and follower. Using graphical methods, find the pressure angle at the position shown.
- 8-5 Figure P8-3 shows a cam and follower. Using graphical methods, find and sketch the equivalent fourbar linkage for this position of the cam and follower.
- 8-6 Figure P8-3 shows a cam and follower. Using graphical methods, find the pressure angle at the position shown.
- †8-7 Design a double-dwell cam to move a follower from 0 to 2.5" in 60° , dwell for 120° , fall 2.5" in 30° and dwell for the remainder. The total cycle must take 4 sec. Choose suitable programs for rise and fall to minimize accelerations. Plot the $s v a j$ diagrams.
- †8-8 Design a double-dwell cam to move a follower from 0 to 1.5" in 45° , dwell for 150° , fall 1.5" in 90° and dwell for the remainder. The total cycle must take 6 sec. Choose suitable programs for rise and fall to minimize velocities. Plot the $s v a j$ diagrams.
- †8-9 Design a single-dwell cam to move a follower from 0 to 2" in 60° , fall 2" in 90° and dwell for the remainder. The total cycle must take 2 sec. Choose suitable programs for rise and fall to minimize accelerations. Plot the $s v a j$ diagrams.
- †8-10 Design a three-dwell cam to move a follower from 0 to 2.5" in 40° , dwell for 100° , fall 1.5" in 90° , dwell for 20° , fall 1" in 30° and dwell for the remainder. The total cycle must take 10 sec. Choose suitable programs for rise and fall to minimize velocities. Plot the $s v a j$ diagrams.
- †8-11 Design a four-dwell cam to move a follower from 0 to 2.5" in 40° , dwell for 100° , fall 1.5" in 90° , dwell for 20° , fall 0.5" in 30° , dwell for 40° , fall 0.5" in 30° and dwell for the remainder. The total cycle must take 15 sec. Choose suitable programs for rise and fall to minimize accelerations. Plot the $s v a j$ diagrams.
- †8-12 Size the cam from Problem 8-7 for a 1" radius roller follower considering pressure angle and radius of curvature. Use eccentricity only if necessary to balance those functions. Plot both those functions. Draw the cam profile. Repeat for a flat-faced follower. Which would you use?
- †8-13 Size the cam from Problem 8-8 for a 1.5" radius roller follower considering pressure angle and radius of curvature. Use eccentricity only if necessary to balance those

† These problems are suited to solution using program DYNACAM, which is on the attached CD-ROM.

functions. Plot both those functions. Draw the cam profile. Repeat for a flat-faced follower. Which would you use?

- †8-14 Size the cam from Problem 8-9 for a 0.5" radius roller follower considering pressure angle and radius of curvature. Use eccentricity only if necessary to balance those functions. Plot both those functions. Draw the cam profile. Repeat for a flat-faced follower. Which would you use?
- †8-15 Size the cam from Problem 8-10 for a 2" radius roller follower considering pressure angle and radius of curvature. Use eccentricity only if necessary to balance those functions. Plot both those functions. Draw the cam profile. Repeat for a flat-faced follower. Which would you use?
- †8-16 Size the cam from Problem 8-11 for a 0.5" radius roller follower considering pressure angle and radius of curvature. Use eccentricity only if necessary to balance those functions. Plot both those functions. Draw the cam profile. Repeat for a flat-faced follower. Which would you use?
- †8-17 A high friction, high inertia load is to be driven. We wish to keep peak velocity low. Combine segments of half period cycloidal displacements with a constant velocity segment on both rise and fall to reduce the maximum velocity below that obtainable with a full period modified sine acceleration alone (i.e., with no constant velocity portion). Rise is 1" in 90°, dwell for 60°, fall in 50°, dwell for remainder. Compare the two designs and comment. Use an ω of one for comparison.
- †8-18 A constant velocity of 0.4 in/sec must be matched for 1.5 sec. Then the follower must return to your choice of start point and dwell for 2 sec. The total cycle time is 6 sec. Design a cam for a follower radius of 0.75" and a maximum pressure angle of 30° absolute value.
- †8-19 A constant velocity of 0.25 in/sec must be matched for 3 sec. Then the follower must return to your choice of start point and dwell for 3 sec. The total cycle time is 12 sec. Design a cam for a follower radius of 1.25" and a maximum pressure angle of 35° absolute value.
- †8-20 A constant velocity of 2 in/sec must be matched for 1 second. Then the follower must return to your choice of start point. The total cycle time is 2.75 sec. Design a cam for a follower radius of 0.5" and a maximum pressure angle of 25° absolute value.
- †8-21 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the s v a j diagrams for a modified trapezoidal acceleration cam function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.
- †8-22 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the s v a j diagrams for a modified sine acceleration cam function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.
- †8-23 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the s v a j diagrams for a cycloidal displacement cam function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.
- †8-24 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the s v a j diagrams for a 3-4-5 polynomial displacement cam

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

‡ These problems are suited to solution using program DYNACAM, which is on the attached CD-ROM..

function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.

- †8-25 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the $s v a j$ diagrams for a 4-5-6-7 polynomial displacement cam function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.
- †8-26 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the $s v a j$ diagrams for a simple harmonic displacement cam function for any specified values of lift and duration. Test it using a lift of 20 mm over 60° at 1 rad/sec.
- †8-27 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a modified trapezoidal acceleration cam function for any specified values of lift, duration, eccentricity, and prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?
- †8-28 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a modified sine acceleration cam function for any specified values of lift, duration, eccentricity, and prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?
- †8-29 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a cycloidal displacement cam function for any specified values of lift, duration, eccentricity, and prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?
- †8-30 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a 3-4-5 polynomial displacement cam function for any specified values of lift, duration, eccentricity, and prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?
- †8-31 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a 4-5-6-7 polynomial displacement cam function for any specified values of lift, duration, eccentricity, and prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?
- †8-32 Write a computer program or use an equation solver such as *Mathcad* or *TKSolver* to calculate and plot the pressure angle and radius of curvature for a simple harmonic displacement cam function for any specified values of lift, duration, eccentricity, and

† These problems are suited to solution using *Mathcad*, *Matlab*, or, *TKSolver* equation solver programs.

prime circle radius. Test it using a lift of 20 mm over 60° at 1 rad/sec, and determine the prime circle radius needed to obtain a maximum pressure angle of 20° . What is the minimum diameter of roller follower needed to avoid undercutting with these data?

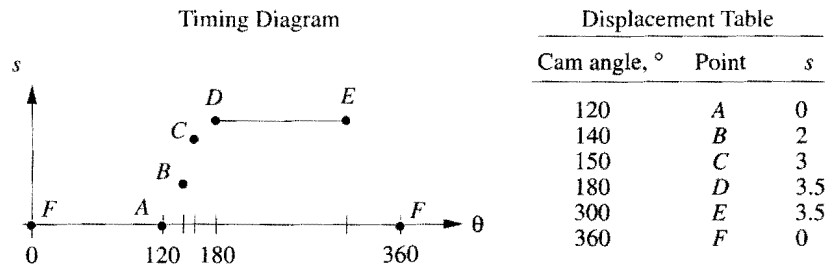
- 8-33 Derive equation 8.21 (p. 382) for the 4-5-6-7 polynomial function.
- 8-34 Derive an expression for the pressure angle of a barrel cam with zero eccentricity.
- ‡8-35 Design a radial plate cam to move a translating roller follower through 30 mm in 30° , dwell for 100° , fall 10 mm in 10° , dwell for 20° , fall 20 mm in 20° , and dwell for the remainder. Camshaft $\omega = 200$ rpm. Minimize the follower's peak velocity and determine the minimum prime circle radius that will give a maximum 25° pressure angle. Determine the minimum radii of curvature on the pitch curve.
- ‡8-36 Repeat Problem 8-35, but minimize the follower's peak acceleration instead.
- ‡8-37 Repeat Problem 8-35, but minimize the follower's peak jerk instead.
- ‡8-38 Design a radial plate cam to lift a translating roller follower through 10 mm in 65° , return to 0 in 65° and dwell for the remainder. Camshaft $\omega = 3500$ rpm. Minimize the cam size while not exceeding a 25° pressure angle. What size roller follower is needed?
- ‡8-39 Design a cam-driven quick-return mechanism for a 3:1 time ratio. The translating roller follower should move forward and back 50 mm and dwell in the back position for 80° . It should take one-third the time to return as to move forward. Camshaft $\omega = 100$ rpm. Minimize the package size while maintaining a 25° maximum pressure angle. Draw a sketch of your design and provide s , v , a , j , ϕ , and p diagrams.
- ‡8-40 Design a cam-follower system to drive a linear translating piston at constant velocity for 200° through a stroke of 100 mm at 60 rpm. Minimize the package size while maintaining a 25° maximum pressure angle. Draw a sketch of your design and provide s , v , a , j , ϕ , and p diagrams.

8.12 PROJECTS

These larger-scale project statements deliberately lack detail and structure and are loosely defined. Thus, they are similar to the kind of "identification of need" or problem statement commonly encountered in engineering practice. It is left to the student to structure the problem through background research and to create a clear goal statement and set of task specifications before attempting to design a solution. This design process is spelled out in Chapter 1 and should be followed in all of these examples. All results should be documented in a professional engineering report. (See the bibliography of Chapter 1 for information on report writing.)

- ‡P8-1 A timing diagram for a halogen headlight filament insertion device is shown in Figure P8-4. Four points are specified. Point A is the start of rise. At B the grippers close to grab the filament from its holder. The filament enters its socket at C and is fully inserted at D. The high dwell from D to E holds the filament stationary while it is soldered in place. The follower returns to its start position from E to F. From F to A the follower is stationary while the next bulb is indexed into position. It is desirable to have low to zero velocity at point B where the grippers close on the fragile

‡ These problems are suited to solution using program DYNACAM, which is on the attached CD-ROM..

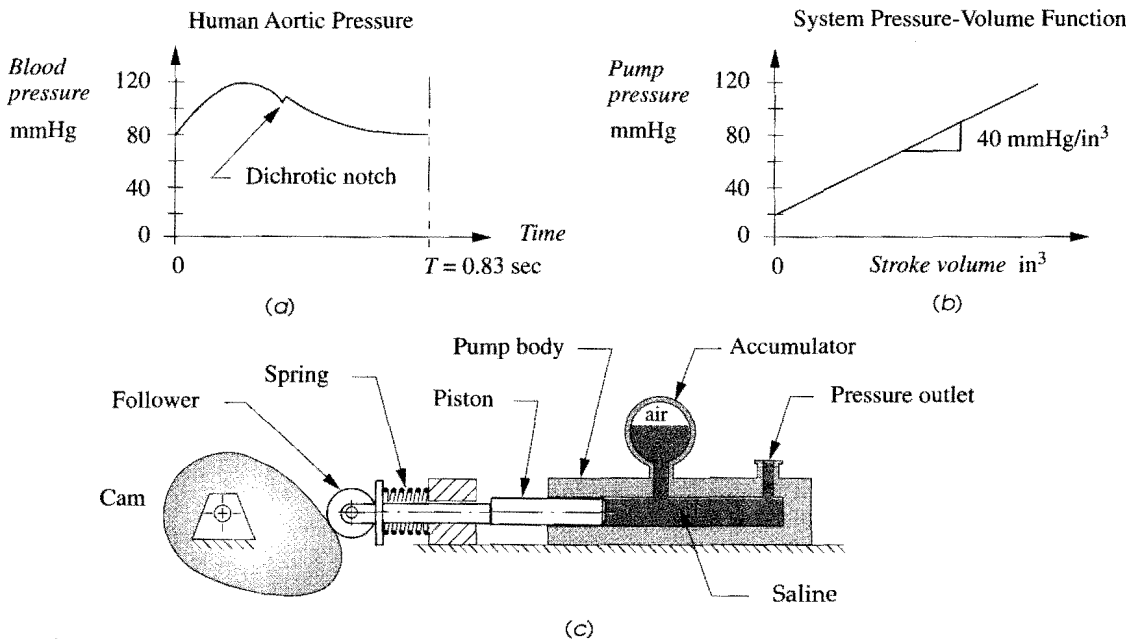
**FIGURE P8-4**

Data for cam design Project P8-1

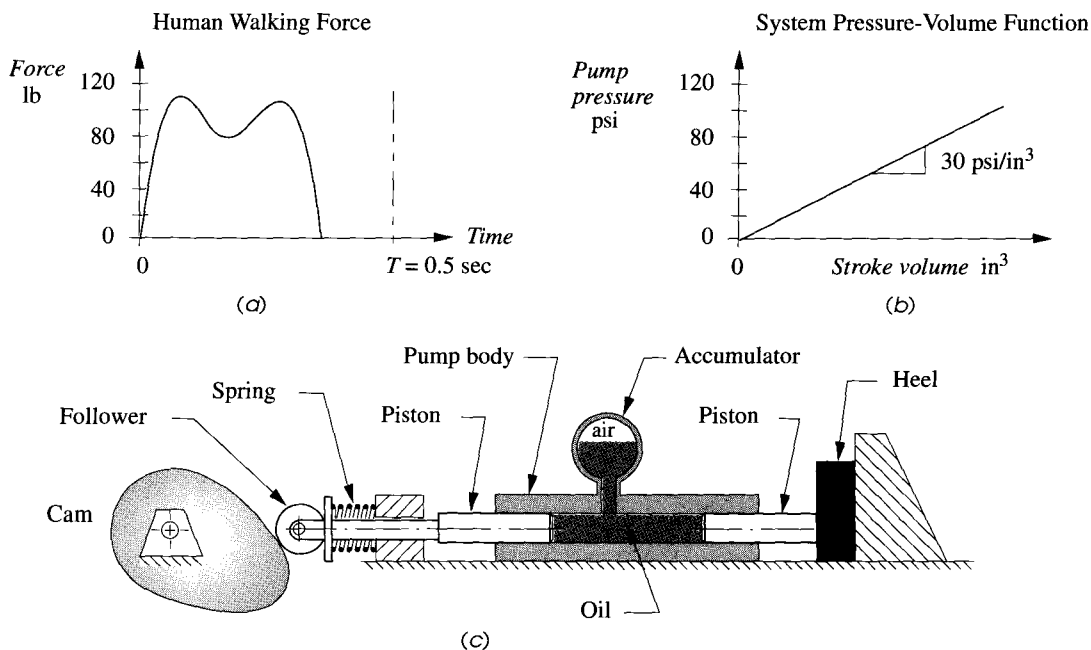
filament. The velocity at C should not be so high as to “bend the filament in the breeze.” Design and size a complete cam-follower system to do this job.

- ‡P8-2 A cam-driven pump to simulate human aortic pressure is needed to serve as a consistent, repeatable pseudo-human input to the operating room computer monitoring equipment, in order to test it daily. Figure P8-5 shows a typical aortic pressure curve and a pump pressure-volume characteristic. Design a cam to drive the piston and give as close an approximation to the aortic pressure curve shown as can be obtained without violating the fundamental law of cam design. Simulate the dichrotic notch as best you can.

- ‡P8-3 An athletic footwear manufacturer wants a device to test rubber heels for their ability to withstand millions of cycles of force similar to that which a walking human’s foot applies to the ground. Figure P8-6 shows a typical walker’s force-time function and a

**FIGURE P8-5**

Data for cam design Project P8-2

**FIGURE P8-6**

Data for cam design Project P8-3

pressure-volume curve for a piston-accumulator. Design a cam-follower system to drive the piston in a way that will create a force-time function on the heel similar to the one shown. Choose suitable piston diameters at each end.

- ‡P8-4 A fluorescent light bulb production machine moves 5500 lamps per hour through a 550°C oven on a chain conveyor which is in constant motion. The lamps are on 2-in centerlines. The bulbs must be sprayed internally with a tin oxide coating as they leave the oven, still hot. This requires a cam-driven device to track the bulbs at constant velocity for the 0.5 sec required to spray them. The spray guns will fit on a 6×10 in table. The spray creates hydrochloric acid, so all exposed parts must be resistant to that environment. The spray head transport device will be driven from the conveyor chain by a shaft having a 28-tooth sprocket in mesh with the chain. Design a complete spray gun transport assembly to these specifications.
- ‡P8-5 A 30-ft-tall drop tower is being used to study the shape of water droplets as they fall through air. A camera is to be carried by a cam-operated linkage which will track the droplet's motion from the 8-ft to the 10-ft point in its fall (measured from release point at the top of the tower). The drops are released every $1/2$ sec. Every drop is to be filmed. Design a cam and linkage which will track these droplets, matching their velocities and accelerations in the 1-ft filming window.
- ‡P8-6 A device is needed to accelerate a 3000-lb vehicle into a barrier with constant velocity, to test its 5 mph bumpers. The vehicle will start at rest, move forward, and have constant velocity for the last part of its motion before striking the barrier with the specified velocity. Design a cam-follower system to do this. The vehicle will leave contact with your follower just prior to the crash.

‡ These problems are suited to solution using program DYNACAM, which is on the attached CD-ROM..

Chapter 9

GEAR TRAINS

*Cycle and epicycle,
orb in orb*

JOHN MILTON, *PARADISE LOST*

9.0 INTRODUCTION

The earliest known reference to gear trains is in a treatise by Hero of Alexandria (c. 100 B.C.). Gear trains are widely used in all kinds of mechanisms and machines, from can openers to aircraft carriers. Whenever a change in the speed or torque of a rotating device is needed, a gear train or one of its cousins, the belt or chain drive mechanism, will usually be used. This chapter will explore the theory of gear tooth action and the design of these ubiquitous devices for motion control. The calculations involved are trivial compared to those for cams or linkages. The shape of gear teeth has become quite standardized for good kinematic reasons which we will explore.

Gears of various sizes and styles are readily available from many manufacturers. Assembled gearboxes for particular ratios are also stock items. The kinematic design of gear trains is principally involved with the selection of appropriate ratios and gear diameters. A complete gear train design will necessarily involve considerations of strength of materials and the complicated stress states to which gear teeth are subjected. This text will not deal with the stress analysis aspects of gear design. There are many texts which do. Some are listed in the bibliography at the end of this chapter. This chapter will discuss the kinematics of gear tooth theory, gear types, and the kinematic design of gearsets and gear trains of simple, compound, reverted, and epicyclic types. Chain and belt drives will also be discussed. Examples of the use of these devices will be presented as well.

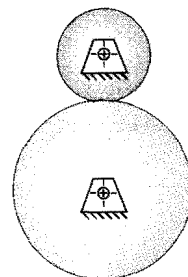
9.1 ROLLING CYLINDERS

The simplest means of transferring rotary motion from one shaft to another is a pair of rolling cylinders. They may be an external set of rolling cylinders as shown in Figure 9-1a or an internal set as in Figure 9-1b. Provided that sufficient friction is available at the rolling interface, this mechanism will work quite well. There will be no slip between the cylinders until the maximum available frictional force at the joint is exceeded by the demands of torque transfer.

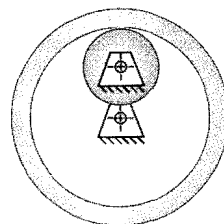
A variation on this mechanism is what causes your car or bicycle to move along the road. Your tire is one rolling cylinder and the road the other (very large radius) one. Friction is all that prevents slip between the two, and it works well unless the friction coefficient is reduced by the presence of ice or other slippery substances. In fact, some early automobiles had rolling cylinder drives inside the transmission, as do some present-day snowblowers and garden tractors which use a rubber-coated wheel rolling against a steel disk to transmit power from the engine to the wheels.

A variant on the rolling cylinder drive is the flat or vee belt as shown in Figure 9-2. This mechanism also transfers power through friction and is capable of quite large power levels, provided enough belt cross section is provided. Friction belts are used in a wide variety of applications from small sewing machines to the alternator drive on your car, to multihorsepower generators and pumps. Whenever absolute phasing is not required and power levels are moderate, a friction belt drive may be the best choice. They are relatively quiet running, require no lubrication, and are inexpensive compared to gears and chain drives.

Both rolling cylinders and belt (or chain) drives have effective linkage equivalents as shown in Figure 9-3. These effective linkages are valid only for one instantaneous position but nevertheless show that these devices are just another variation of the four-bar linkage in disguise.



(a) External set



(b) Internal set

FIGURE 9-1

Rolling cylinders

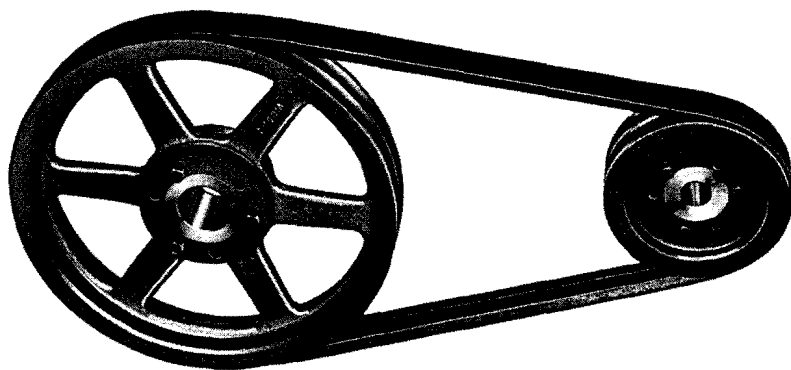


FIGURE 9-2

A two-groove vee belt drive Courtesy of T. B. Wood's Sons Co., Chambersburg, PA

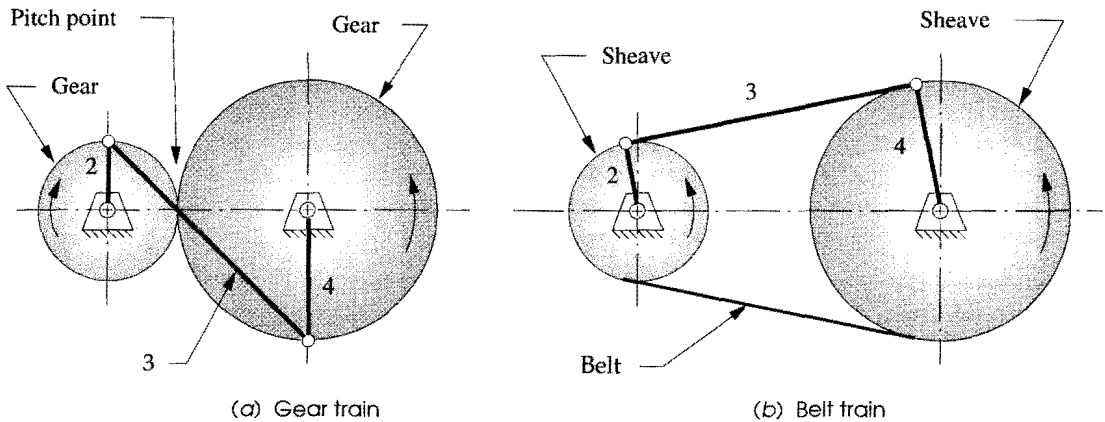


FIGURE 9-3

Gear and belt trains each have an equivalent fourbar linkage for any instantaneous position

The principal drawbacks to the rolling cylinder drive (or smooth belt) mechanism are its relatively low torque capability and the possibility of slip. Some drives require absolute phasing of the input and output shafts for timing purposes. A common example is the valve train drive in an automobile engine. The valve cams must be kept in phase with the piston motion or the engine will not run properly. A smooth belt or rolling cylinder drive from crankshaft to camshaft would not guarantee correct phasing. In this case some means of preventing slip is needed.

This usually means adding some meshing teeth to the rolling cylinders. They then become gears as shown in Figure 9-4 and are together called a *gearset*. When two gears are placed in mesh to form a gearset such as this one, it is conventional to refer to the smaller of the two gears as the *pinion* and to the other as the *gear*.

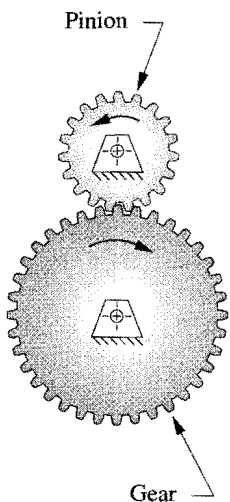


FIGURE 9-4

An external gearset

9.2 THE FUNDAMENTAL LAW OF GEARING

Conceptually, teeth of any shape will prevent gross slip. Old water-powered mills and windmills used wooden gears whose teeth were merely round wooden pegs stuck into the rims of the cylinders. Even ignoring the crudity of construction of these early examples of gearsets, there was no possibility of smooth velocity transmission because the geometry of the tooth “pegs” violated the **fundamental law of gearing** which, if followed, provides that *the angular velocity ratio between the gears of a gearset remains constant throughout the mesh*. A more complete and formal definition of this law is given on p. 436. The angular velocity ratio (m_V) referred to in this law is the same one that we derived for the fourbar linkage in Section 6.4 and equation 6.10 (p. 257). It is equal to the ratio of the radius of the input gear to that of the output gear.

$$m_V = \frac{\omega_{out}}{\omega_{in}} = \pm \frac{r_{in}}{r_{out}} = \pm \frac{d_{in}}{d_{out}} \quad (9.1a)$$

$$m_T = \frac{\omega_{in}}{\omega_{out}} = \pm \frac{r_{out}}{r_{in}} = \pm \frac{d_{out}}{d_{in}} \quad (9.1b)$$

The **torque ratio** (m_T) was shown in equation 6.12 (p. 259) to be the reciprocal of the velocity ratio (m_V); thus a gearset is essentially a device to exchange torque for velocity or vice versa. Since there are no applied forces as in a linkage, but only applied torques on the gears, the **mechanical advantage** m_A of a gearset is equal to its torque ratio m_T . The most common application is to reduce velocity and increase torque to drive heavy loads as in your automobile transmission. Other applications require an increase in velocity, for which a reduction in torque must be accepted. In either case, it is usually desirable to maintain a constant ratio between the gears as they rotate. Any variation in ratio will show up as oscillation in the output velocity and torque even if the input is constant with time.

The radii in equation 9.1 are those of the rolling cylinders to which we are adding the teeth. The positive or negative sign accounts for internal or external cylinder sets as defined in Figure 9-1 (p. 433). An external set reverses the direction of rotation between the cylinders and requires the negative sign. An internal gearset or a belt or chain drive will have the same direction of rotation on input and output shafts and require the positive sign in equation 9.1. The surfaces of the rolling cylinders will become the **pitch circles**, and their diameters the **pitch diameters** of the gears. The contact point between the cylinders lies on the line of centers as shown in Figure 9-3a, and this point is called the **pitch point**.

In order for the fundamental law of gearing to be true, the gear tooth contours on mating teeth must be conjugates of one another. There is an infinite number of possible conjugate pairs that could be used, but only a few curves have seen practical application as gear teeth. The **cycloid** still is used as a tooth form in watches and clocks, but most other gears use the **involute** curve for their shape.

The Involute Tooth Form

The involute is a curve which can be generated by unwrapping a taut string from a cylinder (called the evolute) as shown in Figure 9-5. Note the following about this involute curve:

The string is always tangent to the cylinder.

The center of curvature of the involute is always at the point of tangency of the string with the cylinder.

A tangent to the involute is then always normal to the string, the length of which is the instantaneous radius of curvature of the involute curve.

Figure 9-6 shows two involutes on separate cylinders in contact or “in mesh.” These represent gear teeth. The cylinders from which the strings are unwrapped are called the **base circles** of the respective gears. Note that the base circles are necessarily smaller than the pitch circles, which are at the radii of the original rolling cylinders, r_p and r_g . The gear tooth must project both below and above the rolling cylinder surface (pitch circle) and the *involute only exists outside of the base circle*. The amount of tooth that sticks out above the pitch circle is the **addendum**, shown as a_p and a_g for pinion and gear, respectively. These are equal for standard, full-depth gear teeth.

The geometry at this tooth-tooth interface is similar to that of a cam-follower joint as was defined in Figure 8-40 (p. 399). There is a **common tangent** to both

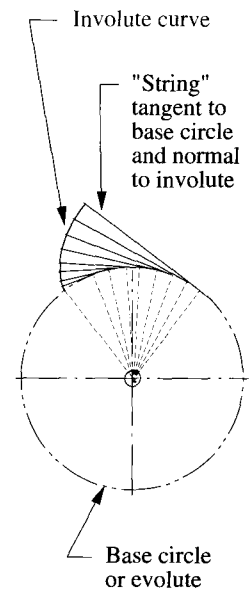


FIGURE 9-5

Development of the involute of a circle

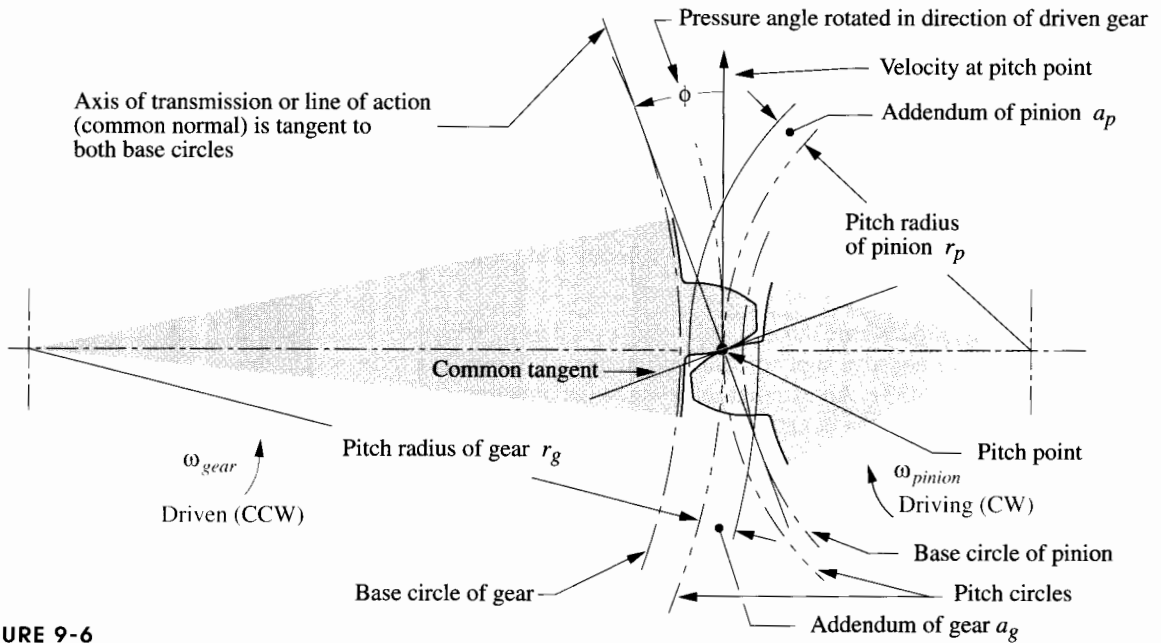


FIGURE 9-6

Contact geometry and pressure angle of involute gear teeth

curves at the contact point, and a **common normal**, perpendicular to the common tangent. Note that the common normal is, in fact, the "strings" of both involutes, which are colinear. Thus the common normal, which is also the **axis of transmission**, always passes through the pitch point regardless of where in the mesh the two teeth are contacting.

Figure 9-7 shows a pair of involute tooth forms in two positions, just beginning contact and about to leave contact. The common normals of both these contact points still pass through the same pitch point. It is this property of the involute that causes it to obey the fundamental law of gearing. The ratio of the driving gear radius to the driven gear radius remains constant as the teeth move into and out of mesh.

From this observation of the behavior of the involute we can restate the fundamental law of gearing in a more kinematically formal way as: *the common normal of the tooth profiles, at all contact points within the mesh, must always pass through a fixed point on the line of centers, called the pitch point.* The gearset's velocity ratio will then be a constant defined by the ratio of the respective radii of the gears to the pitch point.

The points of beginning and leaving contact define the **mesh** of the pinion and gear. The distance along the line of action between these points within the mesh is called the **length of action**, Z , defined by the intersections of the respective addendum circles with the line of action, as shown in Figure 9-7.

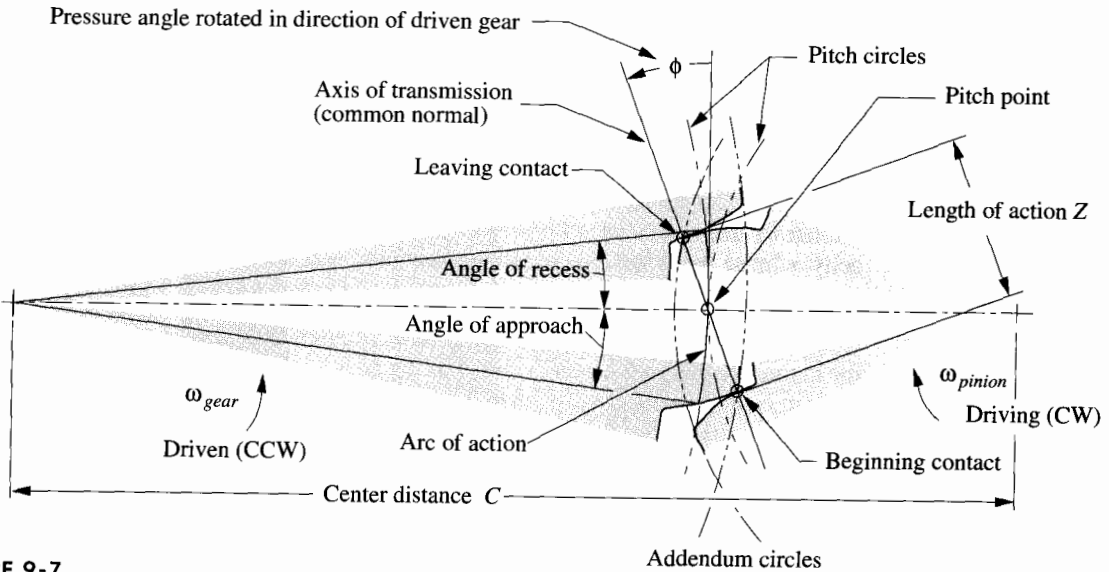


FIGURE 9-7

Pitch point, pitch circles, pressure angle, length of action, arc of action, and angles of approach and recess during the meshing of a gear and pinion

$$Z = \sqrt{(r_p + a_p)^2 - (r_p \cos \phi)^2} + \sqrt{(r_g + a_g)^2 - (r_g \cos \phi)^2} - C \sin \phi \quad (9.2)$$

The distance along the pitch circle within the mesh is the **arc of action**, and the angles subtended by these points and the line of centers are the **angle of approach** and **angle of recess**. These are shown only on the gear in Figure 9-7 for clarity, but similar angles exist for the pinion. The arc of action on both pinion and gear pitch circles must be the same length for zero slip between the theoretical rolling cylinders.

Pressure Angle

The **pressure angle** in a gearset is similar to that of the cam and follower and is defined as the angle between the axis of transmission or line of action (common normal) and the direction of velocity at the pitch point as shown in Figures 9-6 and 9-7. Pressure angles of gearsets are standardized at a few values by the gear manufacturers. These are defined at the nominal center distance for the gearset as cut. The standard values are 14.5°, 20°, and 25° with 20° being the most commonly used and 14.5° now being considered obsolete. Any custom pressure angle can be made, but its expense over the available stock gears with standard pressure angles would be hard to justify. Special cutters would have to be made. Gears to be run together must be cut to the same nominal pressure angle.

Changing Center Distance

When involute teeth (or any teeth) have been cut into a cylinder, with respect to a particular base circle, to create a single gear, we do not yet have a pitch circle. The pitch circle only comes into being when we mate this gear with another to create a pair of gears, or gearset. There will be some range of center-to-center distances over which we can achieve a mesh between the gears. There will also be an ideal center distance (CD) that will give us the nominal pitch diameters for which the gears were designed. However, limitations of manufacturing processes give a low probability that we will be able to exactly achieve this ideal center distance in every case. More likely, there will be some error in the center distance, even if small.

What will happen to the adherence to the fundamental law of gearing if there is error in the location of the gear centers? If the gear tooth form is **not** an involute, then an error in center distance will violate the fundamental law, and there will be variation, or "ripple," in the output velocity. The output angular velocity will not be constant for a constant input velocity. However, **with an involute tooth form, center distance errors do not affect the velocity ratio.** This is the principal advantage of the involute over all other possible tooth forms and the reason why it is nearly universally used for gear teeth. Figure 9-8 shows what happens when the center distance is varied on an involute gearset. Note that the common normal still goes through a pitch point, common to all contact points within the mesh. But the pressure angle is affected by the change in center distance.

Figure 9-8 also shows the pressure angles for two different center distances. As the center distance increases, so will the pressure angle and vice versa. This is one result of a change, or error, in center distance when using involute teeth. Note that the fundamental law of gearing still holds in the modified center distance case. The common normal is still tangent to the two base circles and still goes through the pitch point. The pitch point has moved, but in proportion to the move of the center distance and the gear radii. The velocity ratio is unchanged despite the shift in center distance. In fact, the velocity ratio of involute gears is fixed by the ratio of the base circle diameters, which are unchanging once the gear is cut.

Backlash

Another factor affected by changing center distance is backlash. Increasing the CD will increase the backlash and vice versa. **Backlash** is defined as *the clearance between mating teeth measured along the circumference of the pitch circle*. Manufacturing tolerances preclude a zero clearance, as all teeth cannot be exactly the same dimensions, and all must mesh. So, there must be some small difference between the tooth thickness and the space width (see Figure 9-9). As long as the gearset is run with a nonreversing torque, backlash should not be a problem. But, whenever torque changes sign, the teeth will move from contact on one side to the other. The backlash gap will be traversed and the teeth will impact with noticeable noise. This is the same phenomenon as crossover shock in the form-closed cam. As well as increasing stresses and wear, backlash can cause undesirable positional error in some applications. If the center distance is set exactly to match the theoretical value for the gearset, the tooth-to-tooth composite backlash tolerance is in the range of 0.0001

to 0.0007 in for precision gears. The increase in angular backlash as a function of error in center distance is approximately

$$\theta_B = 43\,200(\Delta C) \frac{\tan \phi}{\pi d} \text{ minutes of arc} \quad (9.3)$$

where ϕ = pressure angle, ΔC = error in center distance, and d = pitch diameter of the gear on the shaft where the backlash is measured.

In servomechanisms, where motors are driving, for example, the control surfaces on an aircraft, backlash can cause potentially destructive “hunting” in which the control system tries in vain to correct positional errors due to backlash “slop” in the mechanical drive system. Such applications need **antibacklash gears** which are really two gears back-to-back on the same shaft which can be rotated slightly at assembly with respect to one another, and then fixed so as to take up the backlash. In less critical applications, such as the propeller drive on a boat, backlash on reversal will not even be noticed.

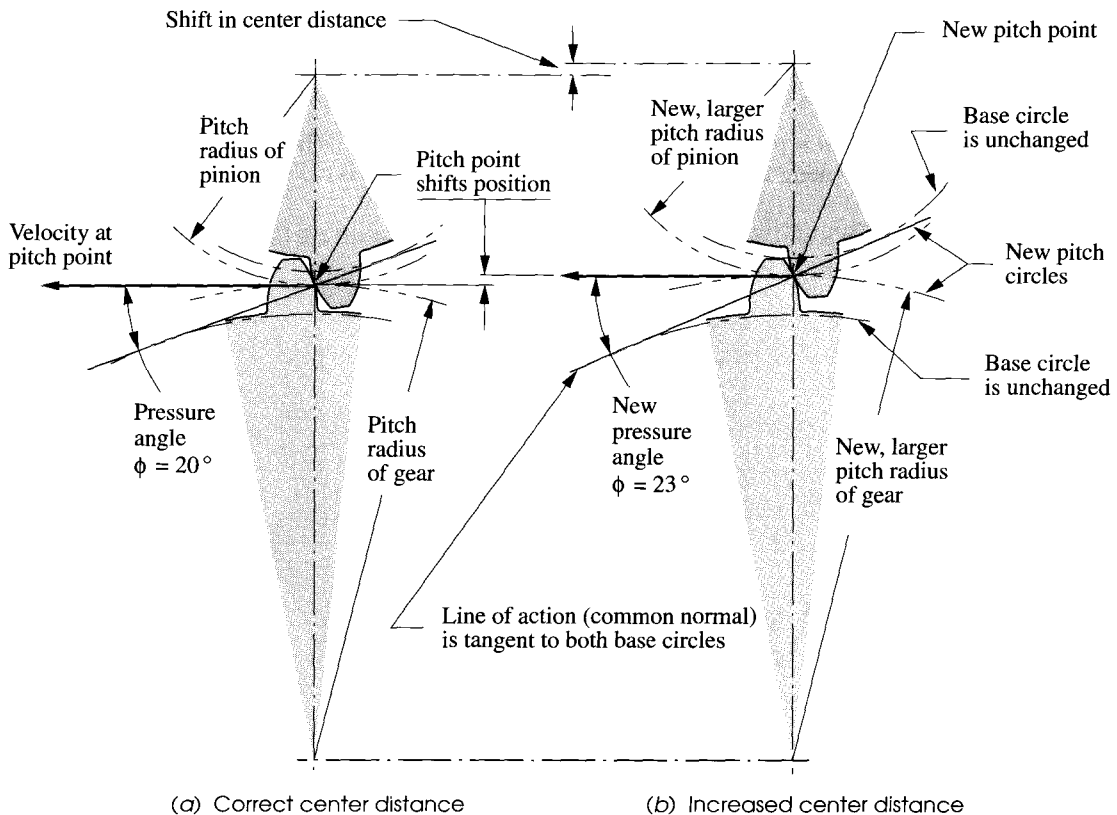


FIGURE 9-8

Changing center distance of involute gears changes the pressure angle and pitch diameters

The *American Gear Manufacturers Association (AGMA)* defines standards for gear design and manufacture. They define a spectrum of quality numbers and tolerances ranging from the lowest (3) to the highest precision (16). Obviously the cost of a gear will be a function of this quality index.

9.3 GEAR TOOTH NOMENCLATURE

Figure 9-9 shows two teeth of a gear with the standard nomenclature defined. **Pitch circle** and **base circle** have been defined above. The tooth height is defined by the **addendum** (*added on*) and the **dedendum** (*subtracted from*) which are referenced to the nominal pitch circle. The dedendum is slightly larger than the addendum to provide a small amount of **clearance** between the tip of one mating tooth (**addendum circle**) and the bottom of the tooth space of the other (**dedendum circle**). The **tooth thickness** is measured at the pitch circle, and the tooth **space width** is slightly larger than the tooth thickness. The difference between these two dimensions is the **backlash**. The **face width** of the tooth is measured along the axis of the gear. The **circular pitch** is the arc length along the pitch circle circumference measured from a point on one tooth to the same point on the next. The circular pitch defines the tooth size. The other tooth dimensions are standardized based on that dimension as shown in Table 9-1. The definition of **circular pitch** p_c is:

$$p_c = \frac{\pi d}{N} \quad (9.4a)$$

where d = pitch diameter and N = number of teeth. The tooth pitch can also be measured along the base circle circumference and then is called the **base pitch** p_b .

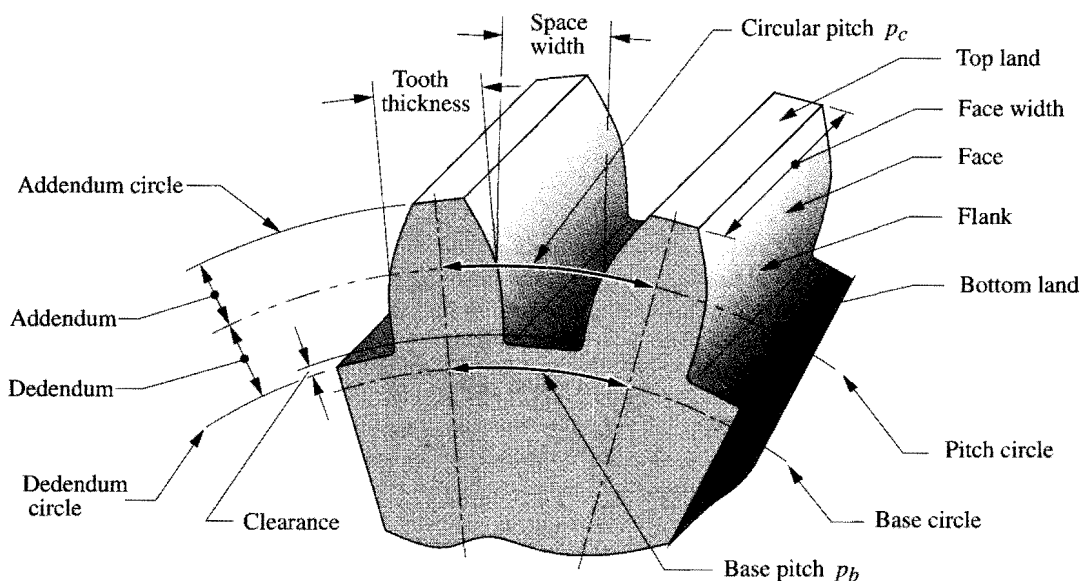


FIGURE 9-9

Gear tooth nomenclature

$$p_b = p_c \cos \phi \quad (9.4b)$$

The units of p_c are inches or millimeters. A more convenient and common way to define tooth size is to relate it to the diameter of the pitch circle rather than its circumference. The **diametral pitch** p_d is:

$$p_d = \frac{N}{d} \quad (9.4c)$$

The units of p_d are reciprocal inches, or number of teeth per inch. This measure is only used in U.S. specification gears. Combining equations 9.4a and 9.4c gives the following relationship between circular pitch and diametral pitch.

$$p_d = \frac{\pi}{p_c} \quad (9.4d)$$

The SI system, used for metric gears, defines a parameter called the **module** which is the reciprocal of diametral pitch with pitch diameter measured in millimeters.

$$m = \frac{d}{N} \quad (9.4e)$$

The units of the module are millimeters. Unfortunately, metric gears are not interchangeable with U.S. gears, despite both being involute tooth forms, as their standards for tooth sizes are different. In the United States, gear tooth sizes are specified by diametral pitch, elsewhere by module. The conversion from one standard to the other is

$$m = \frac{25.4}{p_d} \quad (9.4f)$$

The **velocity ratio** m_V and the **torque ratio** m_T of the gearset can be put into a more convenient form by substituting equation 9.4c into equations 9.1, noting that the diametral pitch of meshing gears must be the same.

TABLE 9-1 AGMA Full-Depth Gear Tooth Specifications

Parameter	Coarse Pitch ($p_d < 20$)	Fine Pitch ($p_d \geq 20$)
Pressure angle ϕ	20° or 25°	20°
Addendum a	1.000 / p_d	1.000 / p_d
Dedendum b	1.250 / p_d	1.250 / p_d
Working depth	2.000 / p_d	2.000 / p_d
Whole depth	2.250 / p_d	2.200 / p_d + 0.002 in
Circular tooth thickness	1.571 / p_d	1.571 / p_d
Fillet radius—basic rack	0.300 / p_d	Not standardized
Minimum basic clearance	0.250 / p_d	0.200 / p_d + 0.002 in
Minimum width of top land	0.250 / p_d	Not standardized
Clearance (shaved or ground teeth)	0.350 / p_d	0.350 / p_d + 0.002 in

$$m_V = \pm \frac{d_{in}}{d_{out}} = \pm \frac{N_{in}}{N_{out}} \quad (9.5a)$$

$$m_T = \pm \frac{d_{out}}{d_{in}} = \pm \frac{N_{out}}{N_{in}} \quad (9.5b)$$

Thus the velocity ratio and torque ratio can be computed from the number of teeth on the meshing gears, which are integers. Note that a minus sign implies an external gearset and a positive sign an internal gearset as shown in Figure 9-1. The gear ratio m_G is always > 1 and can be expressed in terms of either the velocity ratio or torque ratio depending on which is larger than 1. Thus m_G expresses the gear train's overall ratio independent of change in direction of rotation or of the direction of power flow through the train when operated either as a speed reducer or a speed increaser.

$$m_G = |m_V| \text{ or } m_G = |m_T|, \text{ for } m_G \geq 1 \quad (9.5c)$$

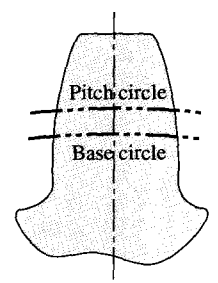
STANDARD GEAR TEETH Standard, full-depth gear teeth have equal addenda on pinion and gear, with the dedendum being slightly larger for clearance. The standard tooth dimensions are defined in terms of the diametral pitch. Table 9-1 (p. 441) shows the definitions of dimensions of standard, full-depth gear teeth as defined by the AGMA, and Figure 9-10 shows their shapes for three standard pressure angles. Figure 9-11 shows the actual sizes of 20° pressure angle, standard, full-depth teeth from $p_d = 4$ to 80. Note the inverse relationship between p_d and tooth size. While there are no theoretical restrictions on the possible values of diametral pitch, a set of standard values is defined based on available gear cutting tools. These standard tooth sizes are shown in Table 9-2 in terms of diametral pitch and in Table 9-3 in terms of metric module.

9.4 INTERFERENCE AND UNDERCUTTING

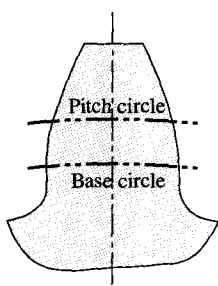
The involute tooth form is only defined outside of the base circle. In some cases, the dedendum will be large enough to extend below the base circle. If so, then the portion of tooth below the base circle will not be an involute and will interfere with the tip of the tooth on the mating gear, which is an involute. If the gear is cut with a standard gear shaper or a "hob," the cutting tool will also interfere with the portion of tooth below the base circle and will cut away the interfering material. This results in an undercut tooth as shown in Figure 9-12. This undercutting weakens the tooth by removing material at its root. The maximum moment and maximum shear from the tooth loaded as a cantilever beam both occur in this region. Severe undercutting will promote early tooth failure.

Interference and its attendant undercutting can be prevented simply by avoiding gears with too few teeth. If a gear has a large number of teeth, they will be small compared to its diameter. As the number of teeth is reduced for a fixed diameter gear, the teeth must become larger. At some point, the dedendum will exceed the radial distance between the base circle and the pitch circle, and interference will occur.

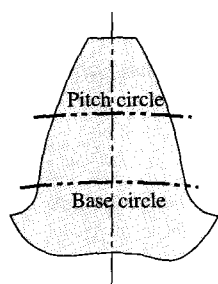
Table 9-4 shows the minimum number of teeth required to avoid undercutting against a standard rack as a function of pressure angle. Table 9-5 shows the minimum number of full-depth pinion teeth that can be used against a selection of full-depth gears of various sizes (for $\phi = 20^\circ$). As the mating gear gets smaller, the pinion can have fewer teeth and still avoid interference.



(a) $\phi = 14.5^\circ$



(b) $\phi = 20^\circ$



(c) $\phi = 25^\circ$

FIGURE 9-10

AGMA full-depth tooth profiles for three pressure angles

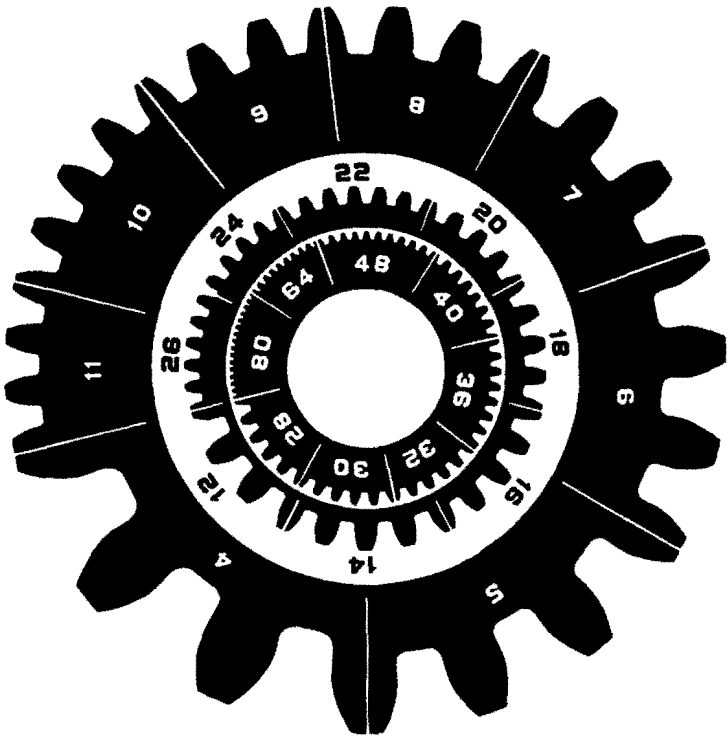


FIGURE 9-11
Actual tooth sizes for various diametral pitches *Courtesy of Barber-Colman Co., Loves Park, IL*

TABLE 9-2 Standard Diametral Pitches	
Coarse ($p_d < 20$)	Fine ($p_d \geq 20$)
1	20
1.25	24
1.5	32
1.75	48
2	64
2.5	72
3	80
4	96
5	120
6	
8	
10	
12	
14	
16	
18	

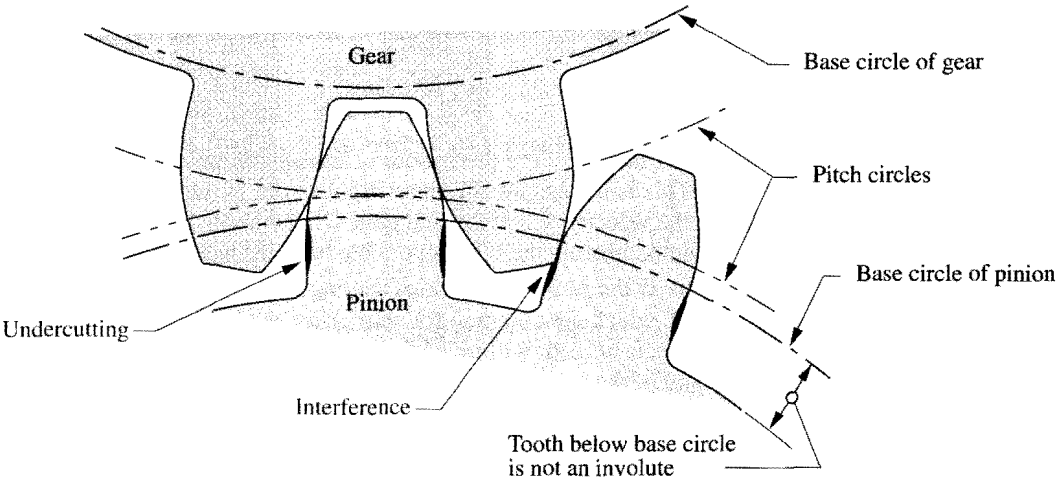


FIGURE 9-12
Interference and undercutting of teeth below the base circle

TABLE 9-3
Standard Metric
Modules

Metric Module (mm)	Equivalent P_d (in^{-1})
0.3	84.67
0.4	63.50
0.5	50.80
0.8	31.75
1	25.40
1.25	20.32
1.5	16.93
2	12.70
3	8.47
4	6.35
5	5.08
6	4.23
8	3.18
10	2.54
12	2.12
16	1.59
20	1.27
25	1.02

Unequal-Addendum Tooth Forms

In order to avoid interference on small pinions, the tooth form can be changed from the standard, full-depth shapes of Figure 9-10 (p. 442) that have equal addenda on both pinion and gear to an involute shape with a longer addendum on the pinion and a shorter one on the gear called **profile-shifted gears**. The AGMA defines addendum modification coefficients, x_1 and x_2 , which always sum to zero, being equal in magnitude and opposite in sign. The positive coefficient x_1 is applied to increase the pinion addendum, and the negative x_2 decreases the gear addendum by the same amount. The total tooth depth remains the same. This shifts the pinion dedendum circle outside its base circle and eliminates that noninvolute portion of pinion tooth below the base circle. The standard coefficients are ± 0.25 and ± 0.50 , which add or subtract 25% or 50% of the standard addendum. The limit of this approach occurs when the pinion tooth becomes pointed.

There are secondary benefits to this technique. The pinion tooth becomes thicker at its base and thus stronger. The gear tooth is correspondingly weakened, but since a full-depth gear tooth is stronger than a full-depth pinion tooth, this shift brings them closer to equal strength. A disadvantage of unequal-addendum tooth forms is an increase in sliding velocity at the tooth tip. The percent sliding between the teeth is greater than with equal addendum teeth which increases tooth-surface stresses. Friction losses in the gear mesh are also increased by higher sliding velocities. Figure 9-13 shows the contours of profile-shifted involute teeth. Compare these to standard tooth shapes in Figure 9-10.

9.5 CONTACT RATIO

The contact ratio m_p defines the average number of teeth in contact at any one time as:

$$m_p = \frac{Z}{p_b} \quad (9.6a)$$

where Z is the length of action from equation 9.2 (p. 437) and p_b is the base pitch from equation 9.4b (p. 441). Substituting equations 9.4b and 9.4d into 9.6a defines m_p in terms of p_d :

$$m_p = \frac{p_d Z}{\pi \cos \phi} \quad (9.6b)$$

If the contact ratio is 1, then one tooth is leaving contact just as the next is beginning contact. This is undesirable because slight errors in the tooth spacing will cause oscillations in the velocity, vibration, and noise. In addition, the load will be applied at the tip of the tooth, creating the largest possible bending moment. At larger contact ratios than 1, there is the possibility of load sharing among the teeth. For contact ratios between 1 and 2, which are common for spur gears, there will still be times during the mesh when one pair of teeth will be taking the entire load. However, these will occur toward the center of the mesh region where the load is applied at a lower position on the tooth, rather than at its tip. This point is called the **highest point of single-tooth contact** (HPSTC). The minimum acceptable contact ratio for smooth operation is 1.2. A minimum contact ratio of 1.4 is preferred and larger is better. Most spur gearsets will have contact ratios between 1.4 and 2. Equation 9.6b shows that for smaller teeth (larger p_d) and larger pressure angle, the contact ratio will be larger.

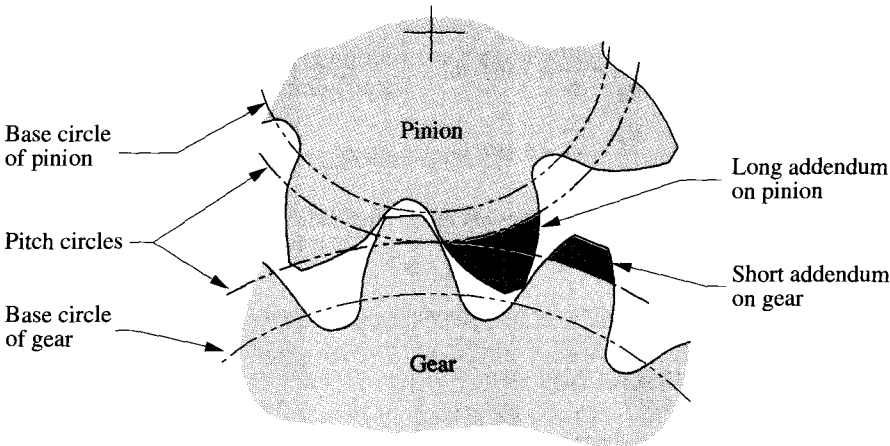


FIGURE 9-13
Profile-shifted teeth with long and short addenda to avoid interference and undercutting

 EXAMPLE 9-1

Determining Gear Tooth and Gear Mesh Parameters.

Problem: Find the gear ratio, circular pitch, base pitch, pitch diameters, pitch radii, center distance, addendum, dedendum, whole depth, clearance, outside diameters, and contact ratio of a gearset with the given parameters. If the center distance is increased 2% what is the new pressure angle and increase in backlash?

Given: A $6 p_d$, 20° pressure angle, 19-tooth pinion is meshed with a 37-tooth gear.

Assume: The tooth forms are standard AGMA full-depth involute profiles.

Solution:

- 1 The gear ratio is found from the tooth numbers on pinion and gear using equation 9.5b (p. 442).

$$m_G = \frac{N_g}{N_p} = \frac{37}{19} = 1.947 \tag{a}$$

- 2 The circular pitch can be found either from equation 9.4a (p. 440) or 9.4c (p. 441).

$$p_c = \frac{\pi}{p_d} = \frac{\pi}{6} = 0.524 \text{ in} \tag{b}$$

- 3 The base pitch measured on the base circle is (from equation 9.4b):

$$p_b = p_c \cos \phi = 0.524 \cos(20^\circ) = 0.492 \text{ in} \tag{c}$$

TABLE 9-4
Minimum Number of Pinion Teeth
To Avoid Interference Between a Full-Depth Pinion and a Full-Depth Rack

Pressure Angle (deg)	Minimum Number of Teeth
14.5	32
20	18
25	12

TABLE 9-5
Minimum Number of Pinion Teeth
To Avoid Interference Between a 20° Full-Depth Pinion and Full-Depth Gears of Various Sizes

Minimum Pinion Teeth	Maximum Gear Teeth
17	1309
16	101
15	45
14	26
13	16

- 4 The pitch diameters and pitch radii of pinion and gear are found from equation 9.4c.

$$d_p = \frac{N_p}{p_d} = \frac{19}{6} = 3.167 \text{ in}, \quad r_p = \frac{d_p}{2} = 1.583 \text{ in} \quad (d)$$

$$d_g = \frac{N_g}{p_d} = \frac{37}{6} = 6.167 \text{ in}, \quad r_g = \frac{d_g}{2} = 3.083 \text{ in} \quad (e)$$

- 5 The nominal center distance C is the sum of the pitch radii:

$$C = r_p + r_g = 4.667 \text{ in} \quad (f)$$

- 6 The addendum and dedendum are found from the equations in Table 9-1 (p. 441):

$$a = \frac{1.0}{p_d} = 0.167 \text{ in}, \quad b = \frac{1.25}{p_d} = 0.208 \text{ in} \quad (g)$$

- 7 The whole depth h_t is the sum of the addendum and dedendum.

$$h_t = a + b = 0.167 + 0.208 = 0.375 \text{ in} \quad (h)$$

- 8 The clearance is the difference between dedendum and addendum.

$$c = b - a = 0.208 - 0.167 = 0.042 \text{ in} \quad (i)$$

- 9 The outside diameter of each gear is the pitch diameter plus two addenda:

$$D_{o_p} = d_p + 2a = 3.500 \text{ in}, \quad D_{o_g} = d_g + 2a = 6.500 \text{ in} \quad (j)$$

- 10 The contact ratio is found from equations 9.2 (p. 437) and 9.6a (p. 444).

$$\begin{aligned} Z &= \sqrt{(r_p + a_p)^2 - (r_p \cos \phi)^2} + \sqrt{(r_g + a_g)^2 - (r_g \cos \phi)^2} - C \sin \phi \\ &= \sqrt{(1.583 + 0.167)^2 - (1.583 \cos 20^\circ)^2} \\ &\quad + \sqrt{(3.083 + 0.167)^2 - (3.083 \cos 20^\circ)^2} - 4.667 \sin 20^\circ = 0.798 \text{ in} \\ m_p &= \frac{Z}{p_b} = \frac{0.798}{0.492} = 1.62 \end{aligned} \quad (k)$$

- 11 If the center distance is increased from the nominal value due to assembly errors or other factors, the effective pitch radii will change by the same percentage. The gears' base radii will remain the same. The new pressure angle can be found from the changed geometry. For a 2% increase in center distance (1.02x):

$$\phi_{new} = \cos^{-1} \left(\frac{r_{\text{base circle } p}}{1.02 r_p} \right) = \cos^{-1} \left(\frac{r_p \cos \phi}{1.02 r_p} \right) = \cos^{-1} \left(\frac{\cos 20^\circ}{1.02} \right) = 22.89^\circ \quad (l)$$

- 12 The change in backlash as measured at the pinion is found from equation 9.3 (p. 439).

$$\theta_B = 43\,200(\Delta C) \frac{\tan \phi}{\pi d} = 43\,200(0.02)(4.667) \frac{\tan(22.89^\circ)}{\pi(3.167)} = 171 \text{ minutes of arc} \quad (m)$$

9.6 GEAR TYPES

Gears are made in many configurations for particular applications. This section describes some of the more common types.

Spur, Helical, and Herringbone Gears

SPUR GEARS are ones in which the *teeth are parallel to the axis of the gear*. This is the simplest and least expensive form of gear to make. Spur gears can only be meshed if their axes are parallel. Figure 9-14 shows a spur gear.

HELICAL GEARS are ones in which the teeth are at a helix angle ψ with respect to the axis of the gear as shown in Figure 9-15a. Figure 9-16 shows a pair of opposite-hand* **helical gears** in mesh. Their axes are parallel. Two **crossed helical gears** of the same hand can be meshed with their axes at an angle as shown in Figure 9-17. The helix angles can be designed to accommodate any skew angle between the nonintersecting shafts.

Helical gears are more expensive than spur gears but offer some advantages. They run quieter than spur gears because of the smoother and more gradual contact between their angled surfaces as the teeth come into mesh. Spur gear teeth mesh along their entire face width at once. The sudden impact of tooth on tooth causes vibrations which are heard as a “whine” which is characteristic of spur gears but is absent with helicals. Also, for the same gear diameter and diametral pitch, a helical gear is stronger due to the slightly thicker tooth form in a plane perpendicular to the axis of rotation.

HERRINGBONE GEARS are formed by joining two helical gears of identical pitch and diameter but of opposite hand on the same shaft. These two sets of teeth are often cut on the same gear blank. The advantage compared to a helical gear is the internal cancellation of its axial thrust loads since each “hand” half of the herringbone gear has an oppositely directed thrust load. Thus no thrust bearings are needed other than to locate the shaft axially. This type of gear is much more expensive than a helical gear and tends to be used in large, high-power applications such as ship drives, where the frictional losses from axial loads would be prohibitive. A herringbone gear is shown in Figure 9-15b. Its face view is the same as the helical gear’s.

EFFICIENCY The general definition of efficiency is *output power/input power* expressed as a percentage. A spur gearset can be 98 to 99% efficient. The helical gearset is less efficient than the spur gearset due to sliding friction along the helix angle. They

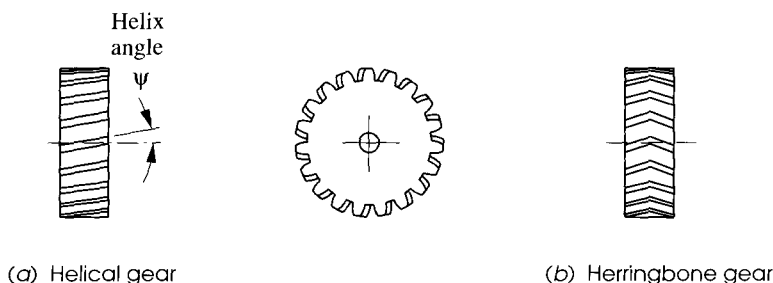


FIGURE 9-15

A helical gear and a herringbone gear

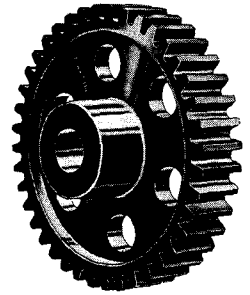


FIGURE 9-14

A spur gear
Courtesy of Martin
Sprocket and Gear Co.,
Arlington, TX

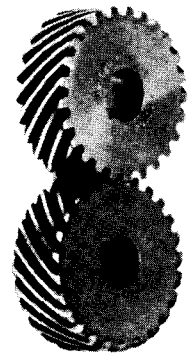


FIGURE 9-16

Parallel axis helical
gears
Courtesy of Martin
Sprocket and Gear Co.,
Arlington, TX

* Helical gears are either right- or left-handed. Note that the gear of Figure 9-15a is left-handed because, if either face of the gear were placed on a horizontal surface, its teeth would slope up to the left.

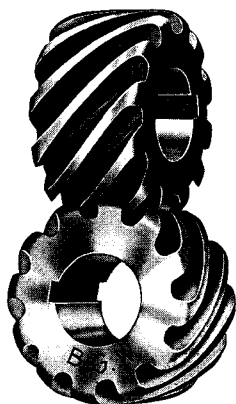


FIGURE 9-17

Crossed axis helical gears

Courtesy of the Boston Gear Division of IMO Industries, Quincy, MA

also present a reaction force along the axis of the gear, which the spur gear does not. Thus helical gearsets must have thrust bearings as well as radial bearings on their shafts to prevent them from pulling apart along the axis. Some friction losses occur in the thrust bearings as well. A parallel helical gearset will be about 96 to 98% efficient, and a crossed helical set only 50 to 90% efficient. The parallel helical set (opposite hand but same helix angle) has line contact between the teeth and can handle high loads at high speeds. The crossed helical set has point contact and a large sliding component which limit its application to light load situations.

If the gearsets have to be shifted in and out of mesh while in motion, spur gears are a better choice than helical, as the helix angle interferes with the axial shifting motion. (Herringbone gears of course cannot be axially disengaged.) Truck transmissions often use spur gears for this reason, whereas automobile (standard) transmissions use helical, constant mesh gears for quiet running and have a synchromesh mechanism to allow shifting. These transmission applications will be described in a later section.

Worms and Worm Gears

If the helix angle is increased sufficiently, the result will be a **worm**, which has only one tooth wrapped continuously around its circumference a number of times, analogous to a screw thread. This worm can be meshed with a special **worm gear** (or **worm wheel**), whose axis is perpendicular to that of the worm as shown in Figure 9-18. Because the driving worm typically has only one tooth, the ratio of the gearset is equal to one over the number of teeth on the worm gear (see equations 9.5, p. 442). These teeth are not involutes over their entire face which means that the center distance must be maintained accurately to guarantee conjugate action.

Worms and wheels are made and replaced as matched sets. These worm gearsets have the advantage of very high gear ratios in a small package and can carry very high loads especially in their single or double enveloping forms. **Single enveloping** means that the worm gear teeth are wrapped around the worm. **Double enveloping** sets also wrap the worm around the gear, resulting in an hourglass-shaped worm. Both of these techniques increase the surface area of contact between worm and wheel, increasing load carrying capacity and also cost. One trade-off in any wormset is very high sliding and thrust loads which make the wormset rather inefficient at 40 to 85% efficiency.

Perhaps the major advantage of the wormset is that it can be designed to be impossible to **backdrive**. A spur or helical gearset can be driven from either shaft, as a velocity step-up or step-down device. While this may be desirable in many cases, if the load being driven must be held in place after the power is shut off, the spur or helical gearset will not do. They will “backdrive.” This makes them unsuitable for such applications as a jack to raise a car unless a brake is added to the design to hold the load. The wormset, on the other hand, can only be driven from the worm. The friction can be large enough to prevent it being backdriven from the worm wheel. Thus it can be used without a brake in load-holding applications such as jacks and hoists.

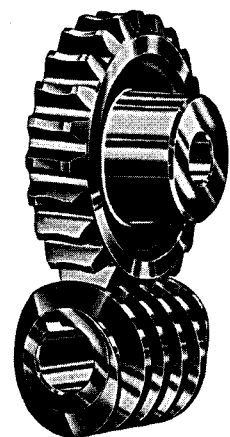


FIGURE 9-18

A worm and worm gear (or worm wheel)

Courtesy of Martin Sprocket and Gear Co., Arlington, TX

Rack and Pinion

If the diameter of the base circle of a gear is increased without limit, the base circle will become a straight line. If the “string” wrapped around this base circle to generate the in-

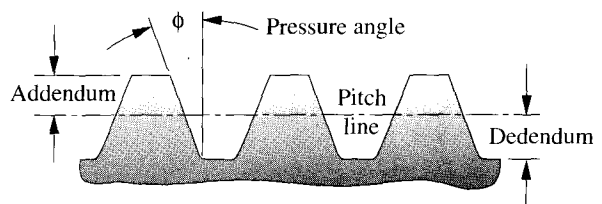
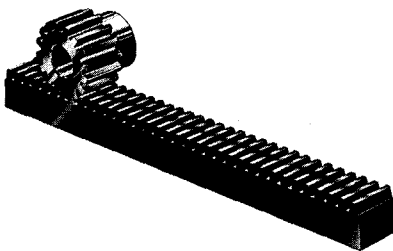
volute were still in place after the base circle's enlargement to an infinite radius, the string would be pivoted at infinity and would generate an involute that is a straight line. This linear gear is called a **rack**. Its teeth are trapezoids, yet are true involutes. This fact makes it easy to create a cutting tool to generate involute teeth on circular gears, by accurately machining a rack and hardening it to cut teeth in other gears. Rotating the gear blank with respect to the rack cutter while moving the cutter axially back and forth across the gear blank will shape, or develop, a true involute tooth on the circular gear.

Figure 9-19 shows a **rack and pinion**. The most common application of this device is in rotary to linear motion conversion or vice versa. It can be backdriven, so it requires a brake if used to hold a load. An example of its use is in **rack-and-pinion steering** in automobiles. The pinion is attached to the bottom end of the steering column and turns with the steering wheel. The rack meshes with the pinion and is free to move left and right in response to your angular input at the steering wheel. The rack is also one link in a multibar linkage which converts the linear translation of the rack to the proper amount of angular motion of a rocker link attached to the front wheel assembly to steer the car.

Bevel and Hypoid Gears

BEVEL GEARS For right-angle drives, crossed helical gears or a wormset can be used. For any angle between the shafts, including 90° , bevel gears may be the solution. Just as spur gears are based on rolling cylinders, **bevel gears** are based on rolling cones as shown in Figure 9-20. The angle between the axes of the cones and the included angles of the cones can be any compatible values as long as the apices of the cones intersect. If they did not intersect, there would be a mismatch of velocity at the interface. The apex of each cone has zero radius, thus zero velocity. All other points on the cone surface will have nonzero velocity. The velocity ratio of the bevel gears is defined by equation 9.1 (p. 434) using the pitch diameters at any common point of intersection of cone diameters.

SPIRAL BEVEL GEARS If the teeth are parallel to the axis of the gear, it will be a straight bevel gear as shown in Figure 9-21. If the teeth are angled with respect to the axis, it will be a **spiral bevel gear** (see Figure 9-22), analogous to a helical gear. The cone axes and apices must intersect in both cases. The advantages and disadvantages of straight bevel and spiral bevel gears are similar to those of the spur gear and helical gear, respectively, regarding strength, quietness, and cost. Bevel gear teeth are not involutes but are based on an "octoid" tooth curve. They must be replaced in pairs (gearsets) as they are not universally interchangeable, and their center distances must be accurately maintained.



Standard full-depth involute rack

FIGURE 9-19

A rack and pinion Photo courtesy of Martin Sprocket and Gear Co., Austin, TX

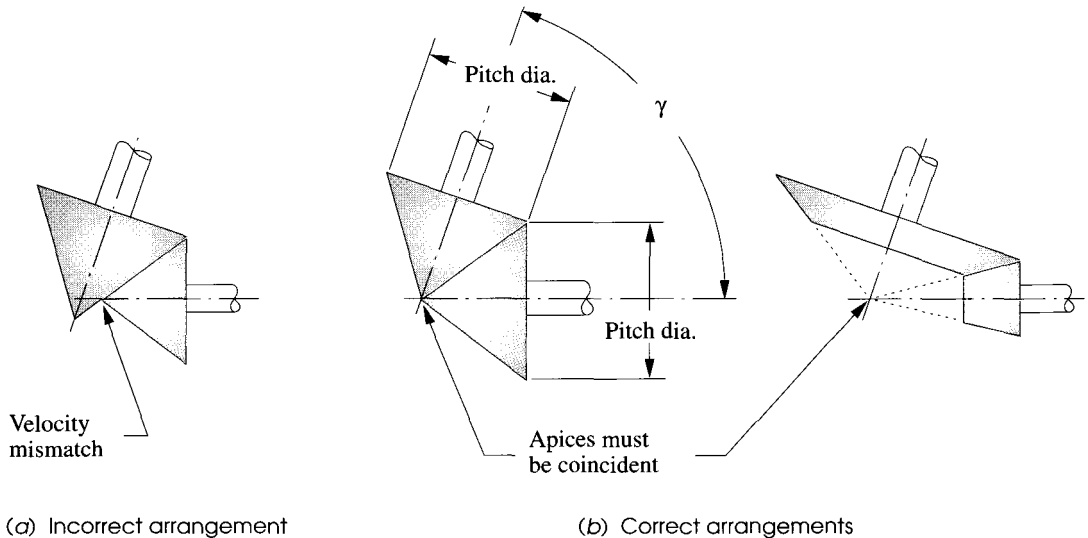


FIGURE 9-20

Bevel gears are based on rolling cones

HVPOIDGEARS If the axes between the gears are nonparallel and also nonintersecting, bevel gears cannot be used. **Hypoid** gears will accommodate this geometry. Hypoid gears are based on rolling hyperboloids of revolution as shown in Figure 9-23. (The term hypoid is a contraction of hyperboloid.) The tooth form is not an involute. These hypoid gears are used in the final drive of front-engine, rear wheel drive automobiles, in order to lower the axis of the driveshaft below the center of the rear axle to reduce the "driveshaft hump" in the back seat.

Noncircular Gears

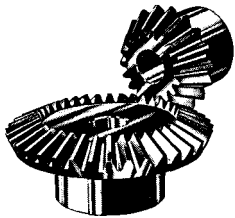


FIGURE 9-21

Straight bevel gears
Courtesy of Martin
Sprocket and Gear,
Arlington, TX

Noncircular gears are based on the rolling centrodes of a Grashof double-crank fourbar linkage. Centrodes are the loci of the instant centers of the linkage and were described in Section 6.5 (p. 263). Figure 6-15b (p. 266) shows a pair of centrodes that could be used for noncircular gears. Teeth would be added to their circumferences in the same way that we add teeth to rolling cylinders for circular gears. The teeth then act to guarantee no slip. Of course, the velocity ratio of noncircular gears is not constant. That is their purpose, to provide a time-varying output function in response to a constant velocity input. Their instantaneous velocity ratio is defined by equation 6.11f (p. 258). These devices are used in a variety of rotating machinery such as printing presses where variation in the angular velocity of rollers is required on a cyclical basis.

Belt and Chain Drives

VEE BELTS A vee belt drive is shown in Figure 9-2 (p. 433). Vee belts are made of elastomers (synthetic rubber) reinforced with synthetic or metallic cords for strength. The pulleys, or *sheaves*, have a matching vee-groove which helps to grip the belt as belt tension

jams the belt into the vee. Flat belts running on flat and crowned pulleys are still used in some applications as well. As discussed above, slip is possible with this device and phasing cannot be guaranteed.

SYNCHRONOUS (TIMING) BELTS The **synchronous belt** solves the phasing problem while retaining the advantages of quiet running of vee belts and can cost less than gears or chains. Figure 9-24a shows a synchronous (or toothed) belt and its special gear-like pulleys or sheaves. These belts are made of a rubberlike material but are reinforced with steel or synthetic cords for higher strength and have molded-in teeth which fit in the grooves of the pulleys for positive drive. They are capable of fairly high torque and power transmission levels and are frequently used to drive automotive engine camshafts as shown in Figure 9-24b. They are more expensive than conventional vee belts. Manufacturers' catalogs provide detailed information on sizing both vee and synchronous belts for various applications.

CHAIN DRIVES are often used for applications where positive drive (phasing) is needed and large torque requirements or high temperature levels preclude the use of timing belts. When the input and output shafts are far apart, a chain drive may be the most economical choice. Conveyor systems often use chain drives to carry the work along the assembly line. Steel chain can be run through many (but not all) hostile chemical or temperature environments. Many types and styles of chain have been designed for various applications ranging from the common roller chain as used on your bicycle or motorcycle, to more expensive "silent chain" designs used for camshaft drives in expensive automobile engines. Figure 9-25 shows a two-row roller chain used for a slightly different task, as a coupling for two in-line shafts. Figure 9-26 shows a typical sprocket for roller chain. Note that the sprocket teeth are not the same shape as gear teeth and are not

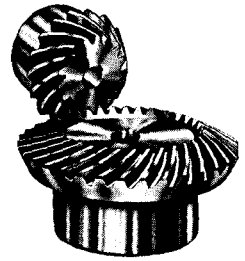
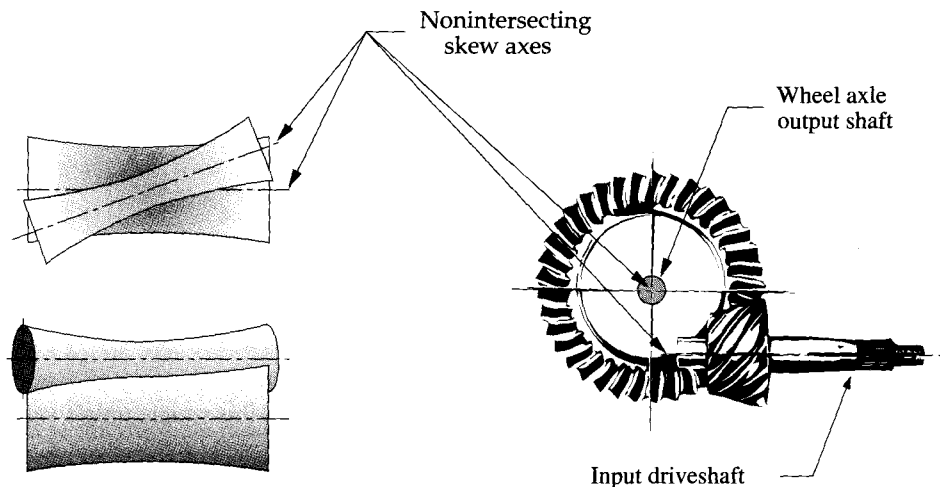


FIGURE 9-22

Spiral bevel gears
Courtesy of the Boston
Gear Division of IMO
Industries, Quincy, MA

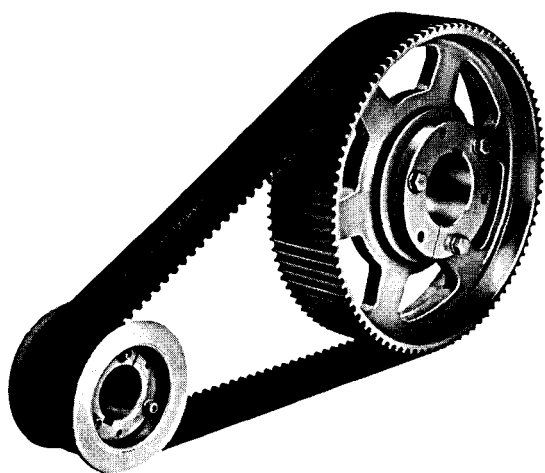


(a) Rolling hyperboloids of revolution

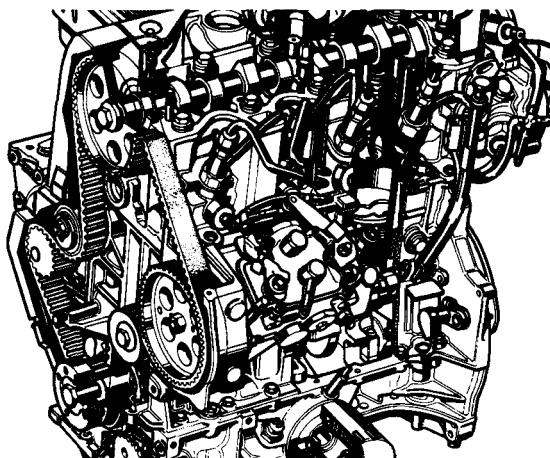
(b) Automotive hypoid final drive gears
Courtesy of General Motors Co., Detroit, MI

FIGURE 9-23

Hypoid gears are based on hyperboloids of revolution



(a) Standard synchronous belt
Courtesy of T. B. Wood's Sons Co.,
Chambersburg, PA



(b) Engine valve camshaft drive
Courtesy of Chevrolet Division,
General Motors Co., Detroit, MI

FIGURE 9-24

Toothed synchronous belts and sprockets

involute. The sprocket tooth shape is dictated by the need to match the contour of the portion of chain which nestles in the grooves. In this case the roller chain has cylindrical pins which engage the sprocket.

One unique limitation of chain drive is something called “**chordal action.**” The links of the chain form a set of chords when wrapped around the circumference of the sprocket. As these links enter and leave the sprocket, they impart a “jerky” motion to the driven shaft which causes some variation, or ripple, on the output velocity. Roller chain does not exactly obey the fundamental law of gearing. If very accurate, constant output velocity is required, a chain drive may not be the best choice.

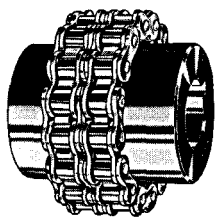


FIGURE 9-25

Roller chain used as a
shaft coupling
Courtesy of Martin
Sprocket and Gear Co.,
Arlington, Texas

9.7 SIMPLE GEAR TRAINS

A gear train is any collection of two or more meshing gears. A simple gear train is one in which each shaft carries only one gear, the most basic, two-gear example of which is shown in Figure 9-4 (p. 434). The *velocity ratio* (sometimes called *train ratio*) of this gearset is found by expanding equation 9.5a (p. 442). Figure 9-27 shows a simple gear train with five gears in series. The expression for this simple train's velocity ratio is:

$$m_V = \left(-\frac{N_2}{N_3} \right) \left(-\frac{N_3}{N_4} \right) \left(-\frac{N_4}{N_5} \right) \left(-\frac{N_5}{N_6} \right) = +\frac{N_2}{N_6}$$

or in general terms:

$$m_V = \pm \frac{N_{in}}{N_{out}} \quad (9.7)$$

which is the same as equation 9.5a for a single gearset.

Each gearset potentially contributes to the overall train ratio, but in any case of a simple (series) train, the numerical effects of all gears except the first and last cancel out. The train ratio of a simple train is always just the ratio of the first gear over the last. Only the sign of the overall ratio is affected by the intermediate gears which are called *idlers* because typically no power is taken from their shafts. If all gears in the train are external and there is an even number of gears in the train, the output direction will be opposite that of the input. If there is an odd number of external gears in the train, the output will be in the same direction as the input. Thus a single, external idler gear of any diameter can be used to change the direction of the output gear without affecting its velocity.

A single gearset of spur, helical, or bevel gears is usually limited to a ratio of about 10:1 simply because the gearset will become very large, expensive, and hard to package above that ratio if the pinion is kept above the minimum numbers of teeth shown in Table 9-4 (p. 445). If the need is to get a larger train ratio than can be obtained with a single gearset, it is clear from equation 9.6 (p. 444) that the simple train will be of no help.

It is common practice to insert a single idler gear to change direction, but more than one idler is superfluous. There is little justification for designing a gear train as is shown in Figure 9-27. If the need is to connect two shafts that are far apart, a simple train of many gears could be used but will be more expensive than a chain or belt drive for the same application. Most gears are not cheap.

9.8 COMPOUND GEAR TRAINS

To get a train ratio of greater than about 10:1 with spur, helical, or bevel gears (or any combination thereof) it is necessary to **compound the train** (unless an epicyclic train is used—see Section 9.9). A **compound train** is one in which at least one shaft carries more than one gear. This will be a parallel or series-parallel arrangement, rather than the pure series connections of the simple gear train. Figure 9-28 shows a compound train of four gears, two of which, gears 3 and 4, are fixed on the same shaft and thus have the same angular velocity.

The train ratio is now:

$$m_V = \left(-\frac{N_2}{N_3} \right) \left(-\frac{N_4}{N_5} \right) \quad (9.8a)$$

This can be generalized for any number of gears in the train as:

$$m_V = \pm \frac{\text{product of number of teeth on driver gears}}{\text{product of number of teeth on driven gears}} \quad (9.8b)$$

Note that these intermediate ratios do not cancel and the overall train ratio is the product of the ratios of parallel gearsets. Thus a larger ratio can be obtained in a compound gear train despite the approximately 10:1 limitation on individual gearset ratios. The plus or minus sign in equation 9.8b depends on the number and type of meshes in the train, whether external or internal. Writing the expression in the form of equation 9.8a and carefully noting the sign of each mesh ratio in the expression will result in the correct algebraic sign for the overall train ratio.

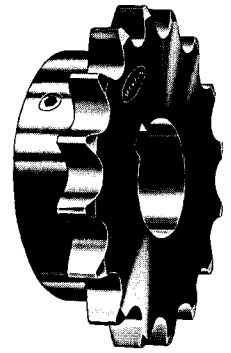


FIGURE 9-26

A roller chain sprocket
Courtesy of Martin
Sprocket and Gear Co.,
Arlington, TX

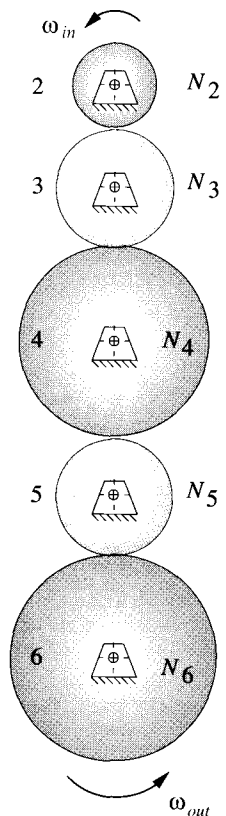


FIGURE 9-27

A simple gear train

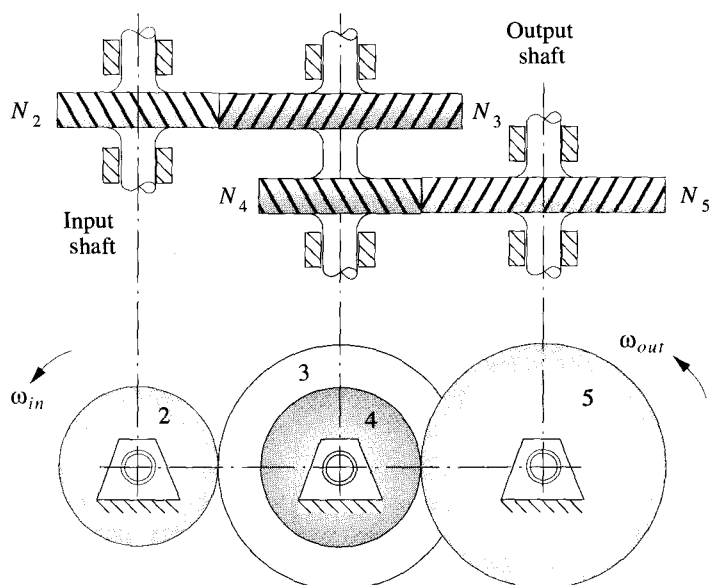


FIGURE 9-28

A compound gear train

Design of Compound Trains

If presented with a completed design of a compound gear train such as that in Figure 9-28, it is a trivial task to apply equation 9.8 and determine the train ratio. It is not so simple to do the inverse, namely, design a compound train for a specified train ratio.

EXAMPLE 9-2

Compound Gear Train Design.

Problem: Design a compound train for an exact train ratio of 180:1. Find a combination of gears which will give that ratio.

Solution:

- 1 The first step is to determine how many stages, or gearsets, are necessary. Simplicity is the mark of good design, so try the smallest possibility first. Take the square root of 180, which is 13.416. So, two stages each of that ratio will give approximately 180:1. However, this is larger than our design limit of 10:1 for each stage, so try three stages. The cube root of 180 is 5.646, well within 10, so three stages will do.
- 2 If we can find some integer ratio of gear teeth which will yield 5.646:1, we can simply use three of them to design our gearbox. Using a lower limit of 12 teeth for the pinion and trying several possibilities we get the gearsets shown in Table 9-6 as possibilities.

TABLE 9-6

Example 9-2

Possible Gearsets for 180:1
Three-Stage Compound
Train

Gearset Ratio	Pinion Teeth	Gear Teeth
$5.646 \times 12 =$	12	67.75
$5.646 \times 13 =$	13	73.40
$5.646 \times 14 =$	14	79.05
$5.646 \times 15 =$	15	84.69

- 3 The number of gear teeth obviously must be an integer. The closest to an integer in Table 9-6 is the 79.05 result. Thus a 79:14 gearset comes closest to the desired ratio. Applying this ratio to all three stages will yield a train ratio of $(79/14)^3 = 179.68:1$, which is within 0.2% of 180:1. This may be an acceptable solution provided that the gearbox is not being used in a timing application. If the purpose of this gearbox is to step down the motor speed for a crane hoist, for example, an approximate ratio will be adequate.
- 4 Many gearboxes are used in production machinery to drive camshafts or linkages from a master driveshaft and must have the exact ratio needed or else the driven device will eventually get out of phase with the rest of the machine. If that were the case in this example, then the solution found above would not be good enough. We will need to redesign it for exactly 180:1. Since our overall train ratio is an integer, it will be simplest to look for integer gear-set ratios. Thus we need three integer factors of 180. The first solution above gives us a reasonable starting point in the cube root of 180, which is 5.65. If we round this up (or down) to an integer, we may be able to find a suitable combination.
- 5 Two compounded stages of 6:1 together give 36:1. Dividing 180 by 36 gives 5. Thus the stages shown in Table 9-7 provide one possible exact solution.

TABLE 9-7
Example 9-2
Exact Solution for 180:1
Three-Stage Compound
Train

Gearset Ratio	Pinion Teeth	Gear Teeth
6 x	14	= 84
6 x	14	= 84
5 x	14	= 70

This solution, shown in Figure 9-29, meets our design criteria. It has the correct, exact ratio; the stages are all less than 10:1; and no pinion has less than 14 teeth, which avoids undercutting if 25° pressure angle gears are used (Table 9-4, p. 445).

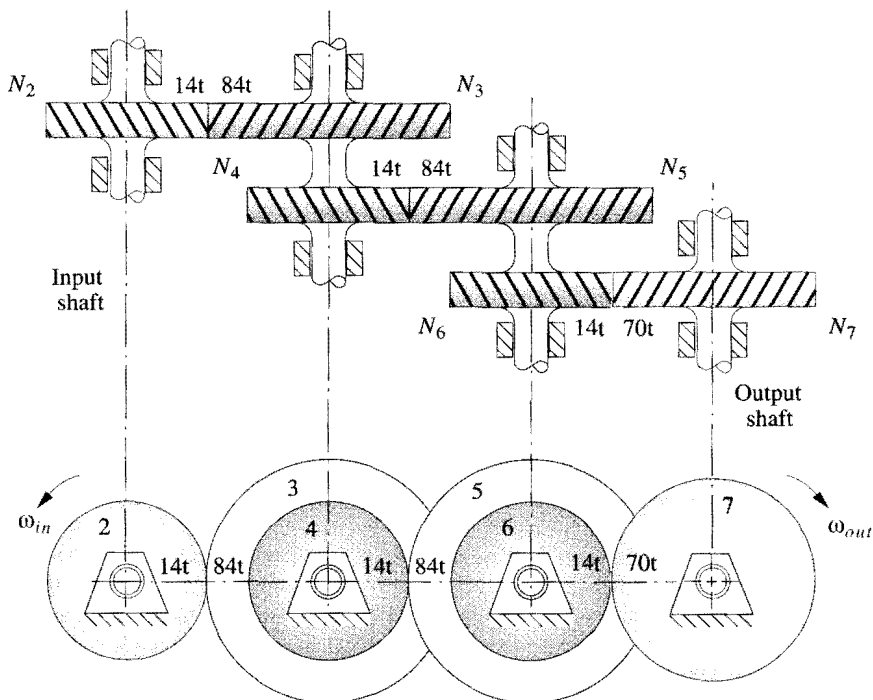


FIGURE 9-29
Three-stage compound gear train for train ratio $m_V = 1:180$ (gear ratio $m_G = 180:1$)

Design of Reverted Compound Trains

In the preceding example the input and output shaft locations are in different locations. This may well be acceptable or even desirable in some cases, depending on other packaging constraints in the overall machine design. Such a gearbox, whose *input and output shafts are not coincident*, is called a **nonreverted compound train**. In some cases, such as automobile transmissions, it is desirable or even necessary to have the *output shaft concentric with the input shaft*. This is referred to as “reverting the train” or “bringing it back onto itself.” The design of a **reverted compound train** is more complicated because of the additional constraint that the center distances of the stages must be equal. Referring to Figure 9-30, this constraint can be expressed in terms of their pitch radii, pitch diameters, or numbers of teeth (provided that all gears have the same diametral pitch).

$$r_2 + r_3 = r_4 + r_5 \quad (9.9a)$$

$$\text{or} \quad d_2 + d_3 = d_4 + d_5 \quad (9.9b)$$

If p_d is the same for all gears, equation 9.4c can be substituted in equation 9.9b and the diametral pitch terms will cancel giving

$$N_2 + N_3 = N_4 + N_5 \quad (9.9c)$$

EXAMPLE 9-3

Reverted Gear Train Design.

Problem: Design a reverted compound train for an exact train ratio of 18:1.

Solution:

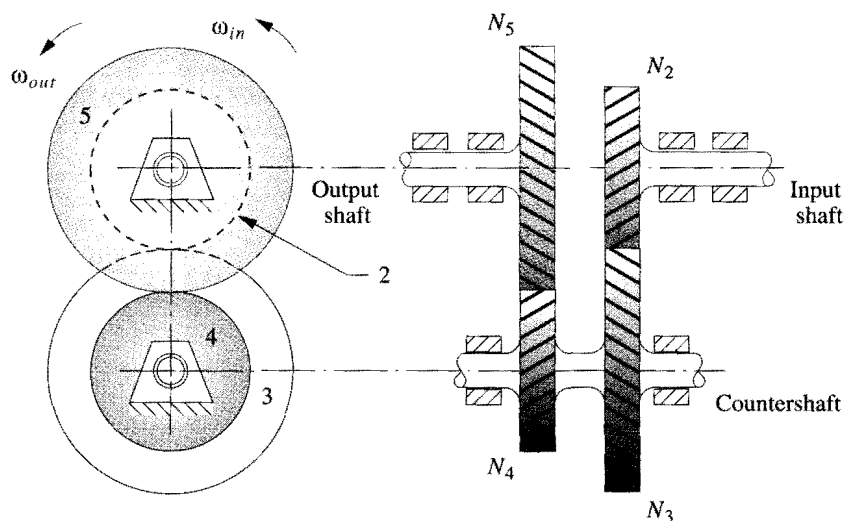
- 1 Though it is not at all necessary to have integer gearset ratios in a compound train (only integer tooth numbers), if the train ratio is an integer, it is easier to design with integer ratio gearsets.
- 2 The square root of 18 is 4.2426, well within our 10:1 limitation. So two stages will suffice in this gearbox.
- 3 If we could form two identical stages, each with a ratio equal to the square root of the overall train ratio, the train would be reverted by default. Table 9-8 shows that there are no reasonable combinations of tooth ratios which will give the exact square root needed. Moreover, this square root is not a rational number, so we cannot get an exact solution by this approach.
- 4 Instead, let's factor the train ratio. All numbers in the factors 9×2 and 6×3 are less than 10, so they are acceptable on that basis. It is probably better to have the ratios of the two stages closer in value to one another for packaging reasons, so the 6×3 choice will be tried.
- 5 Figure 9-30 shows a two-stage reverted train. Note that, unlike the nonreverted train in Figure 9-28 (p. 454), the input and output shafts are now in-line and cantilevered; thus each must have double bearings on one end for moment support and a good bearing ratio as was defined in Section 2.15 (p. 55).

TABLE 9-8

Example 9-3

Possible Gearsets for 18:1
Two-Stage Reverted
Compound Train

Gearset Ratio	Pinion Teeth	Gear Teeth
4.2426	12	50.91
4.2426	13	55.15
4.2426	14	59.40
4.2426	15	63.64
4.2426	16	67.88
4.2426	17	72.12
4.2426	18	76.37
4.2426	19	80.61
4.2426	20	84.85

**FIGURE 9-30**

A reverted compound gear train

- 6 Equation 9.8 states the relationship for its compound train ratio. In addition, we have the constraint that the center distances of the stages must be equal. Use equation 9.9c and set it equal to an arbitrary constant K to be determined.

$$N_2 + N_3 = N_4 + N_5 = K \quad (a)$$

- 7 We wish to solve equations 9.8 and 9.9c simultaneously. We can separate the terms in equation 9.8 and set them each equal to one of the stage ratios chosen for this design.

$$\begin{aligned} \frac{N_2}{N_3} &= \frac{1}{6} \\ N_3 &= 6N_2 \end{aligned} \quad (b)$$

$$\begin{aligned} \frac{N_4}{N_5} &= \frac{1}{3} \\ N_5 &= 3N_4 \end{aligned} \quad (c)$$

- 8 Separating the terms in equation (a):

$$N_2 + N_3 = K \quad (d)$$

$$N_4 + N_5 = K \quad (e)$$

- 9 Substituting equation (b) in (d) and equation (c) in (e) yields:

$$N_2 + 6N_2 = K = 7N_2 \quad (f)$$

$$N_4 + 3N_4 = K = 4N_4 \quad (g)$$

- 10 To make equations (f) and (g) compatible, K must be set to at least the lowest common multiple of 7 and 4, which is 28. This yields values of $N_2 = 4$ teeth and $N_4 = 7$ teeth.
- 11 Since a four-tooth gear will have unacceptable undercutting, we need to increase our value of K sufficiently to make the smallest pinion large enough.
- 12 A new value of $K = 28 \times 3 = 84$ will increase the four-tooth gear to a twelve-tooth gear which is acceptable for a 25° pressure angle (Table 9-4, p. 445). With this assumption of $K = 84$, equations (b), (c), (f), and (g) can be solved simultaneously to give:

$$\begin{array}{ll} N_2 = 12 & N_3 = 72 \\ N_4 = 21 & N_5 = 63 \end{array} \quad (h)$$

which is a viable solution for this reverted train.

The same procedure outlined here can be applied to the design of reverted trains involving several stages such as the helical gearbox in Figure 9-31.

An Algorithm for the Design of Compound Gear Trains

The examples of compound gear train design presented above used integer train ratios. If the required train ratio is noninteger, it is more difficult to find a combination of integer tooth numbers that will give the exact train ratio. Sometimes an irrational gear ratio may be needed for such tasks as conversion of English to metric measure within a machine tool gear train or when π is a factor in the ratio. Then the closest approximation to the desired irrational train ratio that can be contained in a reasonable package is needed.

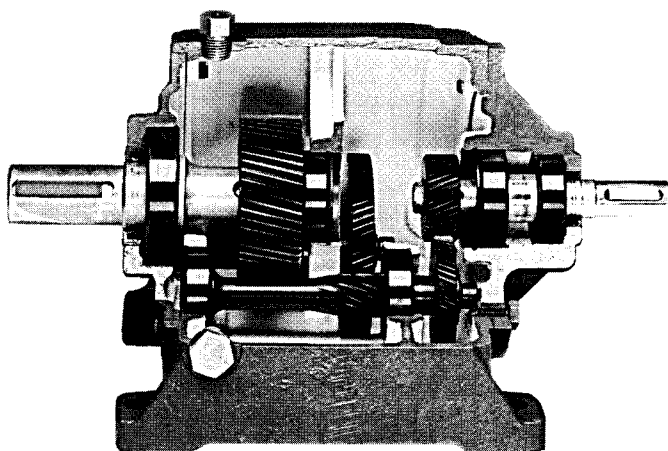


FIGURE 9-31

A commercial, three-stage reverted compound gearbox
 Courtesy of Boston Gear Division of IMO Industries, Quincy, Mass.

DilPare ^[1] and Selfridge and Riddle ^[2] have devised algorithms to solve this problem. Both require a computer for their solution. The Selfridge approach will be described here. It is applicable to two- or three-stage compound trains. A low limit N_{min} and a high limit N_{max} on the acceptable number of teeth for any gear must be specified. An error tolerance ϵ expressed as a percentage of the desired train ratio R (made always > 1) is also selected. For a two-stage compound train the ratio will be as shown in equation 9.5c (p. 442) expanded according to equation 9.8 (p. 453) with the signs neglected for this analysis.

$$R = m_G = \frac{N_3 N_5}{N_2 N_4} \quad (9.10a)$$

The range of acceptable ratios is determined by the choice of error tolerance ϵ .

$$R_{low} = R - \epsilon \quad (9.10b)$$

$$R_{high} = R + \epsilon$$

$$R_{low} \leq \frac{N_3 N_5}{N_2 N_4} \leq R_{high} \quad (9.10c)$$

Then, since the tooth numbers must be integers:

$$N_3 N_5 \leq \text{INT}(N_2 N_4 R_{high}) \quad (9.10d)$$

$$\text{Let :} \quad P = \text{INT}(N_2 N_4 R_{high}) \quad (9.10e)$$

Also from equation 9.10c,

$$N_3 N_5 \geq \text{INT}(N_2 N_4 R_{low}) \quad (9.10f)$$

$$\text{Let :} \quad Q = \text{INT}(N_2 N_4 R_{low}) + 1 \quad (9.10g)$$

rounding up to the next integer.

A search is done on all values of a temporary parameter K defined as $Q \leq K \leq P$ to see if a usable product pair can be found. Because of multiplicative symmetry, the largest value of N_3 that need be considered is

$$N_3 \leq \sqrt{P} \quad (9.11a)$$

$$\text{Let :} \quad N_p = \sqrt{P} \quad (9.11b)$$

The smallest value of N_3 that need be considered occurs when K is at its smallest value Q and N_5 takes its largest value N_{high} . (N_3 is also constrained by N_{low} .)

$$N_3 \geq \frac{Q}{N_{high}} \quad (9.11c)$$

$$\text{Let :} \quad N_m = \text{INT}\left(\frac{Q + N_{high} - 1}{N_{high}}\right) \quad (9.11d)$$

which also rounds up to the next integer.

TABLE 9-9 Algorithm for Design of Two-Stage Compound Gear Trains

From TKSolver file Compound.tk. Based on Reference (2)

" *Ratio* is the desired gear train ratio and must be > 1 . *Nmin* is the minimum number of teeth acceptable on any pinion.
 " *Nmax* is the maximum number of teeth acceptable on any gear. *eps1* is initial estimate of the error tolerance on *Ratio*.
 " *eps* is the tolerance used in the computation, initialized to *eps1* but modified (doubled) until solutions are found.
 " *counter* indicates how many times the initial tolerance was doubled. Note that a large initial value on *eps1* will cause long
 " computation times whereas a too-small value (that gives no solutions) will quickly be increased and lead to a faster solution
 " *pinion1*, *pinion2*, *gear1*, and *gear2* are tooth numbers for solution.

```

    eps = eps1
    counter = 0
redo:
    S = 1
    R_high = Ratio + eps
    R_low = Ratio - eps
    Nh3 = INT( Nmax^2 / R_high )
    Nh4 = INT( Nmax / SQRT( R_high ))
    For pinion1 = Nmin to Nh4
        Nhh = MIN( Nmax, INT( Nh3 / pinion1 ))
        For pinion2 = pinion1 to Nhh
            P = INT( pinion1 * pinion2 * R_high )
            Q = INT( pinion1 * pinion2 * R_low ) + 1
            IF ( P < Q ) THEN GOTO np2
            Nm = MAX( Nmin, INT( (Q + Nmax - 1) / Nmax ))
            Np = SQRT(P)
            For K = Q to P
                For gear1 = Nm to Np
                    IF (MOD( K, gear1 ) <> 0 ) Then GOTO ng1
                    gear2 = K / gear1
                    error = ( Ratio - K / ( pinion1 * pinion2 ) )
                    "check to see if is within current tolerance
                    IF error > eps THEN GOTO ng1
                    " else load solution into arrays
                    pin1[S] = pinion1
                    pin2[S] = pinion2
                    gear1[S] = gear1
                    gear2[S] = gear2
                    error[S] = ABS(error)
                    ratio1[S] = gear1 / pinion1
                    ratio2[S] = gear2 / pinion2
                    ratio[S] = ratio1[S] * ratio2[S]
                    S = S + 1
                Next gear1
            Next K
        Next pinion2
    Next pinion1
    " test to see if any solution occurred with current eps value
    IF (Length( pin1 ) = 0 ) Then GOTO again ELSE Return
again:
    eps = eps * 2
    counter = counter + 1
    GOTO redo

```

" initialize error bound
 " initialize counter
 " reentry point for additional tries at solution
 " initialize the array pointer
 " initialize tolerance bands around ratio
 " initialize tolerance bands around ratio
 " intermediate value for computation
 " intermediate value for computation
 " loop for first pinion
 " intermediate value for computation
 " loop for 2nd pinion
 " intermediate value for computation
 " intermediate value for computation
 " skip to next pinion2 if true
 " intermediate value for computation
 " intermediate value for computation
 " loop for parameter K
 " loop for first gear
 " not a match - skip to next gear1
 " find second gear tooth number
 " find error in ratio
 " is out of bounds - skip to next gear1
 " increment array pointer

The search finds those values of N_3 that meet N_m , N_3 , N_p and $N_5 = K / N_3$. The computer code for this algorithm is shown in Table 9-9. The complete program Compound.tk is on the CD-ROM included with this book, encoded for use with the *TK-Solver* program. The code can be easily rewritten for other equation solvers or compilers.

This algorithm is extendable to three-stage compound gear trains, and the two-stage version can be modified to force reversion of the train by adding a center distance calculation for each gearset and a comparison to a selected tolerance on center distance. These files are also on the CD-ROM as TRIPLE.TK and REVERT.TK, respectively. These programs each generate a table of all solutions that meet the stated error criteria within the tooth limits specified.

EXAMPLE 9-4

Compound Gear Train Design to Approximate an Irrational Ratio.

Problem: Find a pair of gearsets which when compounded will give a train ratio of 3.1416:1 with a minimum error of < 0.001%. Limit gears to tooth numbers between 15 and 100. Also determine the tooth numbers for the smallest error possible if the two gearsets are reverted.

Solution:

- 1 The input data to the algorithm are $R = 3.1416$, $N_{low} = 15$, $N_{high} = 100$, initial $\epsilon = 3.1416E-5$.
- 2 The *TKSolver* file COMPOUND.TK (see Table 9-9) was used to generate the nonreverted solutions shown in Table 9-10.
- 3 The best nonreverted solution has an error in ratio of 7.8499E-06 (0.000 249 87%) giving a ratio of 3.141 582 with gearsets of 29:88 and 85:88 teeth.
- 4 The *TKSolver* file REVERT.TK (see CD-ROM) was used to generate the reverted solutions shown in Table 9-11.

TABLE 9-10 Nonreverted Gearsets and Errors in Ratio for Example 9-4

N_2	N_3	Ratio1	N_4	N_5	Ratio2	m_v	Error
17	54	3.176	91	90	0.989	3.141564	2.5682E-05
17	60	3.529	91	81	0.890	3.141564	2.5682E-05
22	62	2.818	61	68	1.115	3.141580	1.0268E-05
23	75	3.261	82	79	0.963	3.141569	2.0541E-05
25	51	2.040	50	77	1.540	3.141600	1.0000E-05
28	85	3.036	86	89	1.035	3.141611	2.1296E-05
29	88	3.034	85	88	1.035	3.141582	7.8499E-06
33	68	2.061	61	93	1.525	3.141580	1.0268E-05
41	75	1.829	46	79	1.717	3.141569	2.0541E-05
43	85	1.977	56	89	1.589	3.141611	2.1296E-05
43	77	1.791	57	100	1.754	3.141575	1.5133E-05

TABLE 9-11 Reverted Gearsets and Errors in Ratio for Example 9-4

N_2	N_3	<i>Ratio1</i>	N_4	N_5	<i>Ratio2</i>	m_v	<i>Error</i>
22	39	1.773	22	39	1.773	3.142562	-9.6198E-04
44	78	1.773	44	78	1.773	3.142562	-9.6198E-04

- 5 The best reverted solution has an error in ratio of $-9.6198\text{E-}04$ (-0.03062%) giving a ratio of 3.142 562 with gearsets of 22:39 and 22:39 teeth.
- 6 Note that imposing the additional constraint of reversion has reduced the number of possible solutions effectively to one (the two solutions in Table 9-11 differ by a factor of 2 in tooth numbers but have the same error) and the error is much greater than that of even the worst of the 11 nonreverted solutions in Table 9-10.

9.9 EPICYCLIC OR PLANETARY GEAR TRAINS

The conventional gear trains described in the previous sections are all one-degree-of-freedom (*DOF*) devices. Another class of gear train has wide application, the **epicyclic** or **planetary train**. This is a *two-DOF* device. Two inputs are needed to obtain a predictable output. In some cases, such as the automotive differential, one input is provided (the driveshaft) and two frictionally coupled outputs are obtained (the two driving wheels). In other applications such as automatic transmissions, aircraft engine to propeller reductions, and in-hub bicycle transmissions, two inputs are provided (one usually being a zero velocity, i.e., a fixed gear), and one controlled output results.

Figure 9-32a shows a conventional, *one-DOF* gearset in which link 1 is immobilized as the ground link. Figure 9-32b shows the same gearset with link 1 now free to rotate as an **arm** which connects the two gears. Now only the joint 02 is grounded and the system $DOF = 2$. This has become an **epicyclic** train with a **sun gear** and a **planet gear** orbiting around the sun, held in orbit by the arm. Two inputs are required. Typically, the arm and the sun gear will each be driven in some direction at some velocity. In many cases, one of these inputs will be zero velocity, i.e., a brake applied to either the arm or the sun gear. Note that a zero velocity input to the arm merely makes a conventional train out of the epicyclic train as shown in Figure 9-32a. Thus the conventional gear train is simply a special case of the more complex epicyclic train, in which its arm is held stationary.

In this simple example of an epicyclic train, the only gear left to take an output from, after putting inputs to sun and arm, is the planet. It is a bit difficult to get a usable output from this orbiting gear as its pivot is moving. A more useful configuration is shown in Figure 9-33 to which a ring gear has been added. This **ring gear** meshes with the planet and pivots at 02, so it can be easily tapped as the output member. Most planetary trains will be arranged with ring gears to bring the planetary motion back to a grounded pivot. Note how the sun gear, ring gear, and arm are all brought out as concentric hollow shafts so that each can be accessed to tap its angular velocity and torque either as an input or an output.

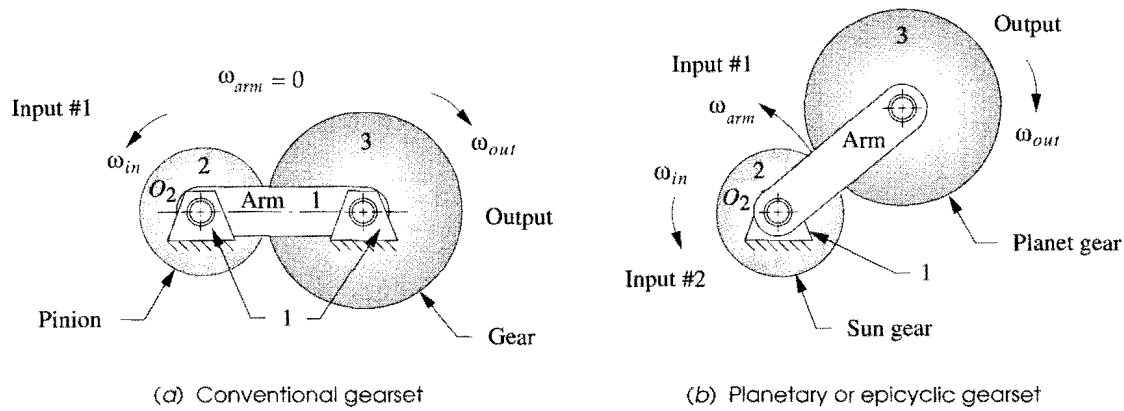


FIGURE 9-32
Conventional gearsets are special cases of planetary or epicyclic gearsets

Epicyclic trains come in many varieties. Levai^[3] catalogued 12 possible types of basic epicyclic trains as shown in Figure 9-34. These basic trains can be connected together to create a larger number of trains having more degrees of freedom. This is done in automotive automatic transmissions as described in a later section.

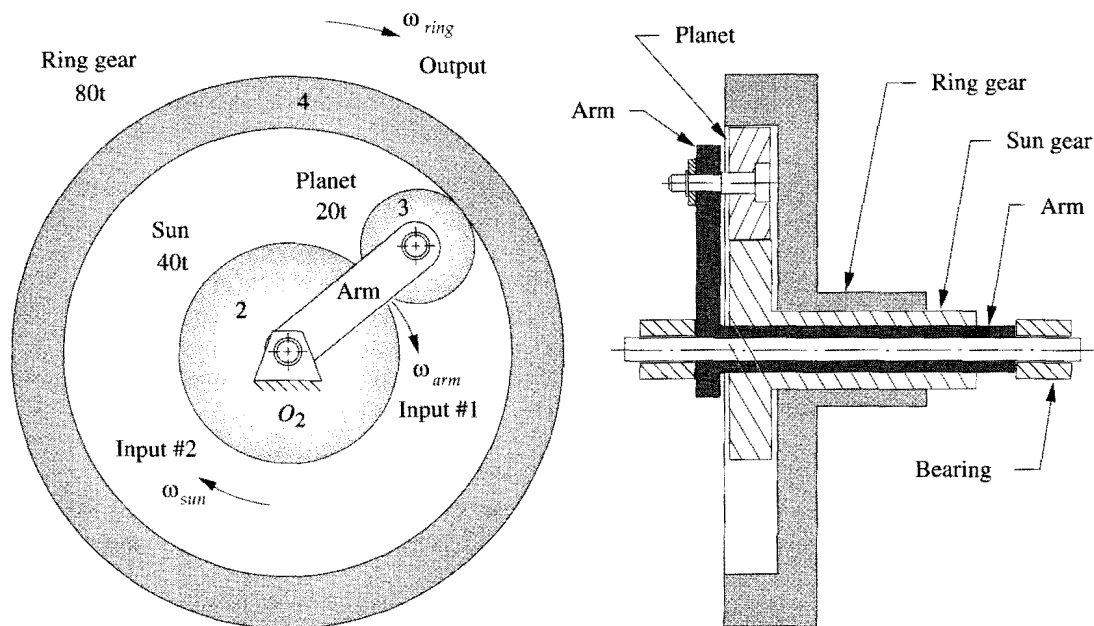


FIGURE 9-33
Planetary gearset with ring gear used as output